

THE EVOLUTION OF ORBITS OF ARTIFICIAL SATELLITES OF PLANETS UNDER THE ACTION OF GRAVITATIONAL PERTURBATIONS OF EXTERNAL BODIES

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Until recently, in works devoted to the evolution of the orbits of artificial satellites, investigations have been made in detail of the influence, on the orbit of the satellite, of the difference of the gravitational field of the Earth and the central and the influence of the braking of the satellite in the Earth's atmosphere. In some works the finer effects of evolution, connected with the rotation of the Earth's atmosphere have also been taken into account. The change in the parameters of the orbits of artificial Earth satellites on account of the gravitational attraction of the Moon and Sun only has been evaluated. Estimations have shown that, when near the Earth, artificial satellites experience the slight influence of other heavenly bodies, which is in practice difficult to observe with present-day means of measurement. However for the American satellite Vanguard I radio-technical means of measurement have already proved to be sufficiently accurate in this respect. The treatment of the results of measurement, in the course of two years of this satellite's existence, has shown⁽¹⁾ that it is impossible to explain the observed evolution of the parameters of the orbit without taking into account the gravitational influence of the Moon and Sun (and also pressure of light).

In the case of the orbit of the satellite Vanguard I, the change of height of the perigee of the orbit, in the course of a year, amounted to a magnitude of the order of 5 km. Such values of evolution are important for accurate forecasting of the position of the satellite, but not suitable for determining the time of its existence and are also obviously unimportant for the carrying out of scientific and technical problems connected with its launching.

In practice in cosmic flights, the important influence of the gravitational attraction of the Moon and Sun on the change of the satellite's orbit was first noted in the case of the Soviet automatic interplanetary station, launched on the 4th October 1959. The orbit of the Earth satellite, into which the automatic interplanetary station turned after nearing the Moon, evolved in such a way^(2,3) that, despite the initial height of perigee of the order of $47 \cdot 10^3$ km, during 11 revolutions the height of perigee, in accordance with the forecast obtained from calculation, became less than the radius of the Earth and the station discontinued its existence; so fundamental may be the effects of the influence of the Moon and Sun for satellites of the type of Explorer VI,⁽⁴⁾ whose orbits when near the surface of the Earth have positions of perigee and heights of apogee $50\text{--}100 \cdot 10^3$ km.

The ratio of the mass of the Earth to the mass of the Moon is equal to $81 \cdot 53$. Owing to this, if we have geometrical equality of the orbits of an Earth satellite and a Moon satellite the latter will experience a perturbation from the Earth $(81 \cdot 53)^2$ times more intensive than the influence of the Moon on such an Earth satellite.

For the study of the evolution of the orbits of artificial satellites, as a rule, we have to investigate a rather wide (in the general case using five measurements) range of possible values of the parameters. For such an investigation the use of exact solutions of systems of differential equations of celestial mechanics, even in the case where the solutions are

obtained with the help of high speed electrical machines, would require expenditure of valuable time and subsequent labour-consuming analysis. Therefore it is quite natural to employ various approximate analytical methods for the investigation. For example, in ⁽⁵⁾, the values of the influence of the Moon and Sun on the initially circular orbit of an artificial Earth satellite were found by the method of obtaining accurate solutions of the linear equations of celestial mechanics. In this work with the simplifications it has been assumed that the ratio of the distance between the position of the satellite in perturbed and unperturbed motion to the radius of the orbit is rather small. The task of the present work was to obtain much simpler formulae for the approximate investigation of the evolution of a sufficiently wide class of satellite orbits. The formulae must, on the one hand, permit us to bring to light the basic qualitative regularities in the character of the changes of the elements of the orbit and, on the other hand, make it possible, with a defined degree of approximation, to obtain quickly quantitative values of the evolution of the orbit for one revolution and for a long period of time. The knowledge of the qualitative regularities permits us to decrease the region of investigation and, with the necessity of a quantitative calculation of evolution over a length of time, approximate formulae markedly shorten the time of calculation.

A basic assumption, used in this work, is the assumption about the rather small value of the ratio of the height of the apocentre of the satellite to the distance from the perturbing body to the central body, around which the satellite is moving. This assumption naturally limits the class of orbit, whose evolution may be considered with the help of the suggested method. However, sacrificing the accuracy of calculation, we may make this class sufficiently wide. Thus, for example, for an Earth satellite with height of apogee $50 \cdot 10^3$ km (approximately $\frac{1}{7}$ of the distance to the Moon), the secular departures of the elements for one revolution of the satellite under the influence of the Moon, according to the approximate formulae obtained below may be calculated with an accuracy of 1–3 per cent.

Other conditions for obtaining approximate formulae for changes of the elements of the orbit for a revolution of the satellite are connected with the small values of these changes. Assumptions of this kind are similar to the assumptions, used in work ⁽⁶⁾, for the approximate calculation of the evolution of the orbit under the action of braking in the atmosphere and as a consequence of the difference between the gravitational field of the Earth and the central. The measurements of the elements of the orbit considered in work ⁽⁶⁾ were either monotonic or periodic functions with large periods. This fact determined the great effectiveness of the different extrapolation methods of calculation of the change in the elements of the orbit over a large interval of time. In the case of gravitational perturbations of the orbit of a satellite by planets, the measurements of the elements of the orbit for a revolution of the satellite, as a function of the number of revolutions, while remaining small in magnitude, have an oscillating character with a frequency which may be comparatively high. The value of this frequency is determined by the period of revolution of the perturbing body around the central one, about which the satellite revolves. This circumstance obviously diminishes the effectiveness of the application, for our problem, of the quicker method used for the accurate calculation of the evolution of the orbits of satellites ⁽⁷⁾, in comparison with their applications to the problems considered in ⁽⁶⁾.

Employing the method of arithmetic averaging of quick-changing coefficients, connected with the motion of the perturbing body, in sections 6 and 7 of this work, formulae are obtained for the approximate calculation of the change in the elements of the orbit of the satellite in a few revolutions or (in certain cases) even for a period of revolution of the perturbing body. This method, in many cases, while not increasing significantly the errors of

the calculation, permit us to speed up the calculations of the change in the elements of the orbit of a satellite over a long period of time.

The equations for the elements of an orbit, averaged throughout a revolution of the perturbing body, on account of their simplicity, are particularly convenient for various kinds of evaluation. For a certain class of orbits and central bodies (it is important that the influence of the non-centrality of the fundamental gravitational field should be a secondary disturbing factor) these equations will describe the main part of the evolution. At the same time, within the limits of such simplified equations, it succeeds in tracing the full evolution for any parameters of the satellite's orbit.

For a given orbit we may estimate the magnitude of the error in calculating, with the approximate formulae, changes in the elements of the satellite's orbit, from the order of the magnitudes which were neglected in the interference of the formulae.

More accurate aggregate values are obviously obtained more simply by comparing the calculation by the approximate formulae with the results of numerical integration of the differential equations, where the arriving at these results with the decrease of the step of integration is a sufficiently reliable criterion of accuracy. In Section 10 the results of such equations for the orbits of satellites of the Earth and Moon are discussed. To obtain the aggregate values of the errors of the approximate method, orbits are considered which are in a certain sense limited, from the standpoint of the application of the approximate method. The errors for these orbits now turn out to be quite appreciable (they consist of magnitudes of a few per cent) but they are obviously permissible for the aims of investigation.

1: THE ORGANIZATION OF THE PROBLEM AND DESIGNATION

We assume that we are given, at some moment of time, the values of the following osculating elements of a satellite, revolving around a central point with constant of gravitational potential μ : p is the parameter of the osculating ellipse; e is its eccentricity; ω the angular distance of the pericentre from the node; i the inclination of its orbit; ϑ the true anomaly; $u = \omega + \vartheta$ the argument of latitude; Ω the longitude of the ascending node.

The reading of the angles i , ω , Ω , u is made from a co-ordinate plane OXY , moving with the central body, belonging to a Cartesian system of co-ordinates XYZ which is not rotating in absolute space.

The movement of the satellite around the central point experiences a perturbation from gravitative points with constants of gravitational potential μ_k ($k = 1, 2, \dots$). The positions of the points relative to the central body are given by the vectors $\mathbf{r}_k(t)$ ($k = 1, 2, \dots$). In the interval of time $t \geq t_0$ considered we investigate a limited problem, the influence of the mass of the satellite on the central body and perturbing bodies being neglected.

In the approximate consideration being carried out we may often consider that the perturbing points are moving along elliptic orbits around the central point. In this case the dependence $\mathbf{r}_k(t)$ is determined by the known elements p_k , e_k , ω_k , Ω_k , i_k , u_k at moment t_0 .

It is assumed that the parameters of the orbit of the satellite and the magnitudes of the perturbing forces are of such a kind that the osculating orbit in the course of one revolution of the satellite in practice differs little from elliptic with fixed parameters, corresponding to the values of the osculating elements, for example, at the pericentre. These elements, fixed for a given revolution, we shall name the elements of the orbit of the satellite.

The change in the elements of the orbit from revolution to revolution gives the secular departure, in consequence of which there may be an important change of form and of disposition in space of the satellite's orbit with respect to the original orbit at moment t_0 .

Digressing from the specific position of the satellite in orbit, from now on we shall be interested only in the aspect of the change of the orbital elements in the course of a definite time.

2. THE INITIAL EQUATIONS AND THE PERTURBING FORCES

To obtain the approximate formulae, describing the evolution of the orbit of an artificial satellite, we shall take initially the exact equations of motion in the osculating elements of⁽⁶⁾ where as an independent variable the value of the true anomaly ϑ is adopted:

$$\left. \begin{aligned} \frac{dp}{d\vartheta} &= \frac{2r^3\gamma}{\mu} T, \\ \frac{de}{d\vartheta} &= \frac{r^2\gamma}{\mu} \left[S \sin \vartheta + \left(1 + \frac{r}{p}\right) T \cos \vartheta + e \frac{r}{p} T \right], \\ \frac{d\omega}{d\vartheta} &= \frac{r^2\gamma}{\mu e} \left[-S \cos \vartheta + \left(1 + \frac{r}{p}\right) T \sin \vartheta - e \frac{r}{p} W \operatorname{ctg} i \sin u \right], \\ \frac{d\Omega}{d\vartheta} &= \frac{r^3\gamma}{\mu p} W \frac{\sin u}{\sin i}, \\ \frac{di}{d\vartheta} &= \frac{r^3\gamma}{\mu p} W \cos u, \end{aligned} \right\} \quad (1)$$

where

$$\gamma = \frac{1}{1 + \frac{r^2}{\mu e} S \cos \vartheta - \frac{r^2}{\mu e} \left(1 + \frac{r}{p}\right) T \sin \vartheta} \quad (2)$$

S , T , W are the projections of the perturbing acceleration respectively on the radius vector, the perpendicular to it in the plane of the osculating ellipse and the perpendicular to the plane of the osculating ellipse.

For a perturbing acceleration $\mathbf{F}^{(k)}$, experienced by the satellite, whose distance from the central body is denoted by the radius vector $\mathbf{r}$ under the action of a gravitating point with potential μ_k , whose distance from the central body is in its turn denoted by the radius vector $\mathbf{r}_k$, the current formula (8) is

$$\mathbf{F}^{(k)} = \mu_k \left(\frac{\mathbf{r}_k - \mathbf{r}}{|\mathbf{r}_k - \mathbf{r}|^3} - \frac{\mathbf{r}_k}{r_k^3} \right). \quad (3)$$

From now on we shall consider satellites, for which the condition $r_k \gg r$ is always fulfilled. In this case the expression for the acceleration $\mathbf{F}^{(k)}$ may be expanded in a series in the param-

eter $\frac{r}{r_k}$

$$\mathbf{F}^{(k)} = \sum_{i=1}^{\infty} \mathbf{F}_i^{(k)}.$$

The first two terms of this series corresponding to the first and second powers of the ratio $\frac{r}{r_k}$,

have the form

$$\mathbf{F}_1^{(k)} = \frac{\mu_k}{r_k^2} \left[3 \frac{\mathbf{r}_k (\mathbf{r}_k \mathbf{r}_k)}{r_k^2} - \frac{\mathbf{r}_k}{r_k} \right], \quad (4)$$

$$\mathbf{F}_2^{(k)} = \frac{\mu_k}{r_k^2} \left[\left(-\frac{3}{2} \frac{r^2}{r_k^2} + \frac{15}{2} \frac{(\mathbf{r}_k \mathbf{r}_k)^2}{r_k^4} \right) \frac{\mathbf{r}_k}{r_k} - 3 \frac{(\mathbf{r}_k \mathbf{r}_k)}{r_k^2} \frac{\mathbf{r}_k}{r_k} \right]. \quad (5)$$

We now introduce into the discussion an orbital system of co-ordinates, in which axis 1 is directed from the central point to the pericentre of the satellite's orbit, axis 2, orthogonal to it, is directed to the side of motion of the body and lies in the plane of the orbit, and axis 3 is directed along the normal to the plane of the satellite's orbit.

The direction cosines of the radius vector $\mathbf{r}_k$, in this system of co-ordinates, we shall call ξ_1, ξ_2, ξ_3 respectively (here and further on we shall omit the index k in the direction cosines).

The expression for the components S, T , and W of the perturbing forces we obtain in the following form:

as a first approximation

$$\left. \begin{aligned} S_1^{(k)} &= \frac{\mu_k}{r_k^2} \frac{r}{r_k} [3\xi^2 \cos^2(\vartheta - \vartheta_\xi) - 1], \\ T_1^{(k)} &= -3 \frac{\mu_k}{r_k^2} \frac{r}{r_k} \xi^2 \cos(\vartheta - \vartheta_\xi) \sin(\vartheta - \vartheta_\xi), \\ W_1^{(k)} &= 3 \frac{\mu_k}{r_k^2} \frac{r}{r_k} \xi_3 \xi \cos(\vartheta - \vartheta_\xi); \end{aligned} \right\} \quad (6)$$

as a second approximation

$$\left. \begin{aligned} S_2^{(k)} &= \frac{15}{2} \frac{\mu_k}{r_k^2} \frac{r^2}{r_k^2} \left[\xi^3 \cos^3(\vartheta - \vartheta_\xi) - \frac{3}{5} \xi \cos(\vartheta - \vartheta_\xi) \right], \\ T_2^{(k)} &= -\frac{15}{2} \frac{\mu_k}{r_k^2} \frac{r^2}{r_k^2} \left[\xi^3 \cos^2(\vartheta - \vartheta_\xi) \sin(\vartheta - \vartheta_\xi) - \frac{1}{5} \xi \sin(\vartheta - \vartheta_\xi) \right], \\ W_2^{(k)} &= \frac{15}{2} \frac{\mu_k}{r_k^2} \frac{r^2}{r_k^2} \xi_3 \left[\xi^2 \cos^2(\vartheta - \vartheta_\xi) - \frac{1}{5} \right]; \end{aligned} \right\} \quad (7)$$

here $\xi = \sqrt{\xi_1^2 + \xi_2^2}$; ϑ_ξ is the true anomaly of the projection of the vector $\mathbf{r}_k$ on the plane of the satellite's orbit, $\sin \vartheta_\xi = \frac{\xi_2}{\xi}$, $\cos \vartheta_\xi = \frac{\xi_1}{\xi}$.

We shall give r_k in the form $r_k = \frac{p_k}{\Delta_k}$, where p_k is the characteristic distance to the perturbing body, Δ_k in the general case a given undimensional function of the time. In the case where the orbit of the perturbing body is an ellipse, p_k the parameter of the orbit and $\Delta_k = 1 + e_k \cos \theta_k$.

Using the relation $r = \frac{p}{\Delta}$, where $\Delta = 1 + e \cos \vartheta$, we may rewrite the expressions for S ,

T , W in the form

$$\left. \begin{aligned} S_1^{(k)} &= 3 \frac{\mu_k}{p_k^2 p_k} \left[\beta_1 \cos^2 \vartheta + 2\beta_3 \cos \vartheta \sin \vartheta - \frac{\beta_6}{3} + \beta_2 \sin^2 \vartheta \right] \frac{1}{\Delta}, \\ T_1^{(k)} &= -3 \frac{\mu_k}{p_k^2 p_k} \left[(\beta_1 - \beta_2) \cos \vartheta \sin \vartheta + \beta_3 (\sin^2 \vartheta - \cos^2 \vartheta) \right] \frac{1}{\Delta}, \\ W_1^{(k)} &= 3 \frac{\mu_k}{p_k^2 p_k} \left[\beta_5 \cos \vartheta + \beta_4 \sin \vartheta \right] \frac{1}{\Delta}, \end{aligned} \right\} \quad (8)$$

$$\left. \begin{aligned} S_2^{(k)} &= \frac{15 \mu_k}{2 p_k^2 p_k^2} \left[\gamma_1 \cos^3 \vartheta + 3\gamma_3 \cos^2 \vartheta \sin \vartheta + 3\gamma_6 \cos \vartheta \sin^2 \vartheta \right. \\ &\quad \left. + \gamma_2 \sin^3 \vartheta - \frac{3}{5} \alpha_1 \cos \vartheta - \frac{3}{5} \alpha_2 \sin \vartheta \right] \frac{1}{\Delta^2}, \\ T_2^{(k)} &= -\frac{15 \mu_k}{2 p_k^2 p_k^2} \left[-\gamma_3 \cos^3 \vartheta + (\gamma_1 - 2\gamma_6) \cos^2 \vartheta \sin \vartheta \right. \\ &\quad \left. + (2\gamma_3 - \gamma_2) \cos \vartheta \sin^2 \vartheta + \gamma_6 \sin^3 \vartheta - \frac{1}{5} \alpha_1 \sin \vartheta + \frac{1}{5} \alpha_2 \cos \vartheta \right] \frac{1}{\Delta^2}, \\ W_2^{(k)} &= \frac{15 \mu_k}{2 p_k^2 p_k^2} \left[\gamma_4 \cos^2 \vartheta + 2\gamma_7 \cos \vartheta \sin \vartheta + \gamma_5 \sin^2 \vartheta - \frac{\alpha_3}{5} \right] \frac{1}{\Delta^2}. \end{aligned} \right\} \quad (9)$$

Here we introduce the designations:

$$\left. \begin{aligned} \alpha_1 &= \xi_1 \Delta_k^4, & \alpha_2 &= \xi_2 \Delta_k^4, & \alpha_3 &= \xi_3 \Delta_k^4, \\ \beta_1 &= \xi_1^2 \Delta_k^3, & \beta_2 &= \xi_2^2 \Delta_k^3, & \beta_3 &= \xi_1 \xi_2 \Delta_k^3, \\ \beta_4 &= \xi_2 \xi_3 \Delta_k^3, & \beta_5 &= \xi_1 \xi_3 \Delta_k^3, & \beta_6 &= \Delta_k^3, \\ \gamma_1 &= \xi_1^3 \Delta_k^4, & \gamma_2 &= \xi_2^3 \Delta_k^4, & \gamma_3 &= \xi_1^2 \xi_2 \Delta_k^4, \\ \gamma_4 &= \xi_1^2 \xi_3 \Delta_k^4, & \gamma_5 &= \xi_2^2 \xi_3 \Delta_k^4, & \gamma_6 &= \xi_2^2 \xi_1 \Delta_k^4, \\ \gamma_7 &= \xi_1 \xi_2 \xi_3 \Delta_k^4. \end{aligned} \right\} \quad (10)$$

3. THE BASIC PRE-REQUISITES FOR OBTAINING THE APPROXIMATE FORMULAE

1. From (2) and (6) it follows that the maximum difference of the value of γ from unity (according to the order of magnitude) is characterized by a parameter proportional to $\frac{\mu_k}{\mu} \left(\frac{a}{r_k} \right)^3 \frac{1}{1-e}$ in the case of eccentricities near unity (here a is the semi-major axis of the orbit). As an example, we shall consider the case of the perturbations by the Moon of an Earth satellite, the semi-major axis of whose orbit is of the order of $30-40 \cdot 10^3$ km. In this case

$$\frac{\mu_k}{\mu} \left(\frac{a}{r_k} \right)^3 \sim 10^{-5}.$$

This value shows that for a wide class of satellite orbit, in practice with all values of the eccentricity, we may, with the defined degree of approximation, take $\gamma = 1$. Further inferences will be based on equations in which $\gamma = 1$.

2. The values of the osculating parameters $p, e, \dots$ for one revolution may be given in the form of a sum of the constant parts of the parameters at some point of the orbit, for example, at its pericentre $p_0, e_0, \dots$ and variable parts $\delta p, \delta e, \dots$

From the equations it follows that the characteristic values of the magnitudes $\delta p, \delta e, \dots$ for a given revolution are proportional to the ratio of the acceleration, called forth by the perturbing body, to the acceleration caused by the central body. Therefore, in the case of orbits sufficiently near the central body, we may neglect the difference between the parameters of the orbit and the constants in the right-hand sides of equations (1), which may lead to a mistake in the determination of the secular departure of the parameters in one revolution of the order $\frac{\delta p}{p}, \frac{\delta e}{e}, \dots$

For example, for an Earth satellite with apogee of orbit of the order $50 \cdot 10^3$ km, this error may be $10^{-3} - 10^{-4}$ of the magnitude of the calculated departure of the elements. Both of the above-mentioned simplifications are standard methods of linearizing equations with small parameters, which have already been used in (6) for the investigations of the evolution of the orbit of a satellite under the action of braking in the atmosphere and on account of the difference between the gravitational field of the earth and the central.

In the approximate consideration adopted, the equations describing the process are linear with linear dependence on the perturbing forces. Therefore, for the determination of the secular departure of the elements for one revolution of the satellite under the action of a sum of accelerations $\sum_{i=1}^n \mathbf{F}_i$, caused by very different perturbing factors, we may determine independently the departures under the influence of each of the factors, and obtain a general result by the simple summation $\Delta p = \sum \Delta p_i, \Delta e = \sum \Delta e_i, \dots$. In particular we may also consider as independent the departures of the elements under the action of each term of the series (3) of the expansion of the perturbing acceleration in powers of $\frac{r}{r_k}$.

3. The change, in the course of time of one revolution, of the magnitudes $\alpha_i, \beta_i, \gamma_i$, determined by the formulae (10), for fixed elements of the orbit, is connected with the motion of the perturbing body in absolute space.

A third important assumption, which will be used for the conclusions of the approximate formulae, is the assumption that the quantities $\alpha_i, \beta_i, \gamma_i$ may be given in the form of a series

$$\left. \begin{aligned} \alpha_i &= \alpha_i^* + \left(\frac{d\alpha_i}{dt}\right)^* \Delta t + \frac{1}{2} \left(\frac{d^2\alpha_i}{dt^2}\right)^* (\Delta t)^2 + \dots, \\ \beta_i &= \beta_i^* + \left(\frac{d\beta_i}{dt}\right)^* \Delta t + \frac{1}{2} \left(\frac{d^2\beta_i}{dt^2}\right)^* (\Delta t)^2 + \dots, \\ \gamma_i &= \gamma_i^* + \left(\frac{d\gamma_i}{dt}\right)^* \Delta t + \frac{1}{2} \left(\frac{d^2\gamma_i}{dt^2}\right)^* (\Delta t)^2 + \dots, \end{aligned} \right\} \quad (11)$$

and that, in an interval of time equal to half the period of revolution of the satellite, they may, with sufficient degree of approximation for the investigation of approximate values of $\alpha_i, \beta_i, \gamma_i$, be limited to a small number of terms of the series (in particular, in the present work to two). Here $\Delta t = t - t^*$, where t^* is some fixed moment of the time interval, corresponding to the given revolution.

4. FORMULAE FOR THE CHANGE OF ELEMENTS OF AN ORBIT FOR ONE REVOLUTION OF THE SATELLITE

We shall substitute in equation (1) the expressions for the perturbing accelerations (8) and integrate with respect to the true anomaly from 0 to 2π . In the result we shall obtain integral formulae for the change of elements for one revolution $\Delta_1^{(k)}p, \Delta_1^{(k)}e, \Delta_1^{(k)}\omega, \Delta_1^{(k)}\Omega, \Delta_1^{(k)}i$. The index k corresponds to the k -th perturbing body, and the lower index shows the approximation used in the expansion of the perturbing acceleration.

The substitution in equations (1) of the second approximation for the perturbing accelerations (9) and the integration with respect to θ from 0 to 2π gives integral formulae for the increments $\Delta_2^{(k)}p, \Delta_2^{(k)}e, \dots$.

For example, for the increment $\Delta_i^{(k)}p$, using the simplifying assumptions of paragraphs 1 and 2 of the approximate discussion of section 3, we have

$$\left. \begin{aligned} \Delta_1^{(k)}p &= -6 \frac{\mu_k p^4}{\mu p_k^3} \left[E \left([\beta_1 - \beta_2] \frac{\cos \vartheta \sin \vartheta}{\Delta^4} \right) + E \left(\beta_3 \frac{\sin^2 \vartheta}{\Delta^4} \right) - E \left(\beta_3 \frac{\cos^2 \vartheta}{\Delta^4} \right) \right], \\ \Delta_2^{(k)}p &= -15 \frac{\mu_k p}{\mu} \left(\frac{p}{p_k} \right)^4 \left[-E \left(\gamma_3 \frac{\cos^3 \vartheta}{\Delta^5} \right) + E \left([\gamma_1 - 2\gamma_6] \frac{\cos^2 \vartheta \sin \vartheta}{\Delta^5} \right) \right. \\ &\quad + E \left([2\gamma_3 - \gamma_2] \frac{\cos \vartheta \sin^2 \vartheta}{\Delta^5} \right) + E \left(\gamma_6 \frac{\sin^3 \vartheta}{\Delta^5} \right) - \frac{1}{5} E \left(\alpha_1 \frac{\sin \vartheta}{\Delta^5} \right) \\ &\quad \left. + \frac{1}{5} E \left(\alpha_2 \frac{\cos \vartheta}{\Delta^5} \right) \right], \end{aligned} \right\} \quad (12)$$

where

$$E[\beta_j f(\vartheta)] = \int_0^{2\pi} \beta_j f(\vartheta) d\vartheta.$$

Similar formulae (a little more cumbersome) are obtained also for the change of other elements of the orbit.

All functions $f(\vartheta)$, entering into the integrals, are rational trigonometrical functions of the type

$$f(\vartheta) = \frac{\cos^m \vartheta \sin^l \vartheta}{(1 + e \cos \vartheta)^n}. \quad (13)$$

Now we use the third assumption of Section 3. Resolving $\alpha_j, \beta_j, \gamma_j$ in series of powers of Δt , we shall obtain the integrals $E(\alpha_j f(\vartheta)), E(\beta_j f(\vartheta)), \dots$ in the form of series of the type

$$E(\beta_j f(\vartheta)) = \beta_j^* E(f(\vartheta)) + \left(\frac{d\beta_j}{dt} \right) E(\Delta t f(\vartheta)) + \frac{1}{2} \left(\frac{d^2 \beta_j}{dt^2} \right)^* E((\Delta t)^2 f(\vartheta)) + \dots \quad (14)$$

Here, as before, $\Delta t = t - t^*$, naturally taking for time t^* the time corresponding to the true anomaly $\vartheta = \pi$, i.e. the time corresponding to the position of the satellite at apocentre.

Accordingly we obtain each of the increments of the elements in the form of a sum

$\Delta_i^{(k)}x = \sum_{s=1}^{\infty} \Delta_{is}^{(k)}x$. For example for $\Delta_1^{(k)}p, \Delta_2^{(k)}p$ we have

$$\left. \begin{aligned} \Delta_i^{(k)}p &= \Delta_{11}^{(k)}p + \Delta_{12}^{(k)}p + \Delta_{13}^{(k)}p + \dots, \\ \Delta_2^{(k)}p &= \Delta_{21}^{(k)}p + \Delta_{22}^{(k)}p + \Delta_{23}^{(k)}p + \dots \end{aligned} \right\} \quad (15)$$

For the first two terms of the first series (15) according to (12) and (14), we shall obtain the following integral formulae:

$$\begin{aligned}\Delta_{11}^{(k)}p &= -6 \frac{\mu_k}{\mu} \frac{p_4}{p_k^3} \left[(\beta_1^* - \beta_2^*) E \left(\frac{\cos \vartheta \sin \vartheta}{\Delta^4} \right) + \beta_3^* E \left(\frac{\sin^2 \vartheta}{\Delta^4} \right) - \beta_3^* E \left(\frac{\cos^2 \vartheta}{\Delta^4} \right) \right], \\ \Delta_{12}^{(k)}p &= -6 \frac{\mu_k}{\mu} \frac{p^4}{p_k^3} \left\{ \left[\left(\frac{d\beta_1}{dt} \right)^* - \left(\frac{d\beta_2}{dt} \right)^* \right] E \left(\Delta t \frac{\cos \vartheta \sin \vartheta}{\Delta^4} \right) \right. \\ &\quad \left. + \left(\frac{d\beta_3}{dt} \right)^* E \left(\Delta t \frac{\sin^2 \vartheta}{\Delta^4} \right) - \left(\frac{d\beta_3}{dt} \right)^* E \left(\Delta t \frac{\cos^2 \vartheta}{\Delta^4} \right) \right\},\end{aligned}$$

$\beta_i^* \left(\frac{d\beta_i}{dt} \right)^*$ here being taken at the moment of time $t = t^*$.

To obtain the final formulae for each constituent increment $\Delta_{ij}^{(k)}x$ of any of the elements of the orbit for a revolution of the satellite, it is necessary to calculate the definite integrals of the form $E \left((\Delta t)^s \frac{\cos^m \vartheta \sin^l \vartheta}{(1 + e \cos \vartheta)^n} \right)$, where s, l, m, n are natural numbers. If as t^* we take the moment when $\theta = \pi$, Δt will be an odd function of θ and the integrals of an odd sum $s + l$ will be equal to zero. For $s = 0$ for the calculation of such integrals there exist recurrent formulae. In the case where the sum $s + l$ is even and $s > 0$, for the calculation of the integrals it is advisable to use a method of integrating by parts and the relation

$$r^2 \frac{d\vartheta}{dt} = \sqrt{p\mu}, \quad (16)$$

which remains correct in the case of the approximate consideration, with the assumption adopted in paragraph 2 of Section 3.

Below come formulae for the increments $\Delta_{11}x$, $\Delta_{12}x$, $\Delta_{21}x$ obtained after the corresponding transformations.

$$\left. \begin{aligned}\Delta_{11}a &= 0, \\ \Delta_{11}e &= -15\pi \frac{\mu_k}{\mu} \left(\frac{a}{p_k} \right)^3 e \varepsilon^{\frac{1}{2}} \beta_3, \\ \Delta_{11}i &= 15\pi \frac{\mu_k}{\mu} \left(\frac{a}{p_k} \right) \frac{1}{\varepsilon^{\frac{1}{2}}} \left[\left(1 - \frac{4}{5} \varepsilon \right) \beta_5 \cos \omega - \frac{1}{5} \varepsilon \beta_4 \sin \omega, \right. \\ \Delta_{11}\Omega &= 15\pi \frac{\mu_k}{\mu} \left(\frac{a}{p_k} \right)^3 \frac{1}{\varepsilon^{\frac{1}{2}} \sin i} \left[\left(1 - \frac{4}{5} \varepsilon \right) \beta_5 \sin \omega + \frac{1}{5} \varepsilon \beta_4 \cos \omega \right], \\ \Delta_{11}\omega &= 3\pi \frac{\mu_k}{\mu} \left(\frac{a}{p_k} \right)^3 \varepsilon^{\frac{1}{2}} [4\beta_1 - \beta_2 - \beta_6] - \Delta_{11}\Omega \cos i,\end{aligned} \right\} \quad (17)$$

$\varepsilon = 1 - e^2$, $a = \frac{p}{\varepsilon}$ is the semi-major axis of the satellite's orbit.

$$\left. \begin{aligned} \Delta_{12}a &= -\frac{3}{2} \frac{\mu_k}{\mu} \left(\frac{a}{p_k}\right)^3 aT \left[e(e+4) \left(\frac{d\beta_2}{dt} - \frac{1}{3} \frac{d\beta_6}{dt} \right) + (2e^2 + 4e - 1) \left(\frac{d\beta_1}{dt} - \frac{d\beta_2}{dt} \right) \right], \\ \Delta_{12}e &= -3 \frac{\mu_k}{\mu} \left(\frac{a}{p_k}\right)^3 \varepsilon T \left[\frac{e+4}{4} \left(\frac{d\beta_2}{dt} - \frac{1}{3} \frac{d\beta_6}{dt} \right) + \left(\frac{1}{8}e - \frac{1}{3} \right) \left(\frac{d\beta_1}{dt} - \frac{d\beta_2}{dt} \right) \right], \\ \Delta_{12}i &= \frac{9}{8} \frac{\mu_k}{\mu} \left(\frac{a}{p_k}\right)^3 T \left(e^2 + \frac{32}{9}e - \frac{2}{3} \right) \left[\frac{d\beta_4}{dt} \cos \omega - \frac{d\beta_5}{dt} \sin \omega \right], \\ \Delta_{12}\Omega &= \frac{9}{8} \frac{\mu_k}{\mu} \left(\frac{a}{p_k}\right)^3 \frac{T}{\sin i} \left(e^2 + \frac{32}{9}e - \frac{2}{3} \right) \left[\frac{d\beta_5}{dt} \cos \omega + \frac{d\beta_4}{dt} \sin \omega \right], \\ \Delta_{12}\omega &= -3 \frac{\mu_k}{\mu} \left(\frac{a}{p_k}\right)^3 \frac{T}{e} \left(\frac{9}{8}e^3 + \frac{8}{3}e^2 - e - \frac{4}{3} \right) \frac{d\beta_3}{dt} - \cos i \Delta_{12}\Omega, \end{aligned} \right\} \quad (18)$$

where $T = \frac{2\pi}{\sqrt{\mu}} a^3$ is the period of revolution of the satellite.

$$\left. \begin{aligned} \Delta_{21}a &= 0, \\ \Delta_{21}e &= \frac{525}{8} \pi \frac{\mu_k}{\mu} \left(\frac{a}{p_k}\right)^4 \varepsilon^{\frac{1}{2}} \left[\left(1 - \frac{6}{7}\varepsilon \right) \gamma_3 + \frac{\varepsilon}{7} \gamma_2 - \left(1 - \frac{3}{7}\varepsilon \right) \frac{\alpha_2}{5} \right], \\ \Delta_{21}i &= -\frac{525}{8} \pi \frac{\mu_k}{\mu} \left(\frac{a}{p_k}\right)^4 \frac{e}{\varepsilon^{\frac{1}{2}}} \left[\left(1 - \frac{4}{7}\varepsilon \right) \gamma_4 \cos \omega \right. \\ &\quad \left. + \frac{1}{7} \varepsilon (\gamma_5 \cos \omega - 2\gamma_7 \sin \omega) - \left(1 - \frac{3}{7}\varepsilon \right) \frac{1}{5} \alpha_3 \cos \omega \right], \\ \Delta_{21}\Omega &= -\frac{525}{8} \pi \frac{\mu_k}{\mu} \left(\frac{a}{p_k}\right)^4 \frac{e}{\varepsilon^{\frac{1}{2}} \sin i} \left[\left(1 - \frac{4}{7}\varepsilon \right) \gamma_4 \sin \omega \right. \\ &\quad \left. + \frac{1}{7} \varepsilon (\gamma_5 \sin \omega + 2\gamma_7 \cos \omega) - \left(1 - \frac{3}{7}\varepsilon \right) \frac{1}{5} \alpha_3 \sin \omega \right], \\ \Delta_{21}\omega &= -\frac{75}{2} \pi \frac{\mu_k}{\mu} \left(\frac{a}{p_k}\right)^4 \frac{e}{\varepsilon^{\frac{1}{2}}} \left[\frac{5}{4} \left(1 - \frac{4}{5}\varepsilon \right) \gamma_4 - \frac{1}{2} \left(1 - \frac{3}{2}\varepsilon \right) \gamma_6 \right. \\ &\quad \left. - \frac{13}{20} \left(1 - \frac{9}{13}\varepsilon \right) \alpha_1 \right] - \Delta_{21}\Omega \cos i. \end{aligned} \right\} \quad (19)$$

Here and from now on the values α_i^* , β_i^* , γ_i^* , $\left(\frac{d\beta_i}{dt}\right)^*$ have the asterisk * omitted. We shall point to some investigations resulting from the formulae obtained above.

1. The equating to zero of the increments $\Delta_{11}a$ and $\Delta_{21}a$ is connected with the fact that the increments $\Delta_{11}x$ and $\Delta_{21}x$ are related to the case of a perturbing point which is motionless in absolute space. For a motionless perturbing body the motion of the satellite takes place in a conservative field of force. So long as it is also assumed that the elements of the orbit do not change in the course of one revolution then, towards the end of the revolution, the satellite is shown to be at the same point in space and, consequently, the increment in the total energy will be zero.

In addition, the increments $\Delta_{i1}p$, $\Delta_{i1}i$, $\Delta_{i1}\Omega$ are connected by the additional relation, the stipulated existence, in the case of a motionless perturbing body, of an integral connected with the conservation of the projection of the vector of angular momentum on the direction of the perturbing body.

In our notation this integral has the form

$$\sqrt{\mu p} \xi_3 = \text{const}, \quad (20)$$

whence $\frac{1}{2} \frac{\Delta_{i1}p}{p} \xi_3 + \Delta_{i1}\xi_3 = 0$. Using the condition $a = \text{const}$, we have $\frac{\Delta_{i1}p}{p} = -2 \frac{e\Delta_{i1}e}{\varepsilon}$. For $\Delta_{i1}\xi_3$ we may obtain the formulae

$$\Delta_{i1}\xi_3 = \bar{\xi}_1 \Delta_{i1}\Omega \sin i - \bar{\xi}_2 \Delta_{i1}i,$$

where

$$\bar{\xi}_1 = \xi_1 \cos \omega - \xi_2 \sin \omega, \quad \bar{\xi}_2 = \xi_1 \sin \omega + \xi_2 \cos \omega.$$

Finally the relation connecting the increments $\Delta_{i1}e$, $\Delta_{i1}\Omega$, $\Delta_{i1}i$, have the form

$$-\frac{e\Delta_{i1}e}{\varepsilon} \xi_3 + (\xi_1 \cos \omega - \xi_2 \sin \omega) \Delta_{i1}\Omega \sin i - (\xi_1 \sin \omega + \xi_2 \cos \omega) \Delta_{i1}i = 0. \quad (21)$$

With the direct substitution of the increments successively from (17) and (19) in (21) we may make certain of the correctness of the integral (20). We may avail ourselves of the relation (21) to check the correctness of the calculation of the increment $\Delta_{i1}x$.

2. In the majority of problems for which the approximate method will be used, the basic qualitative and quantitative effects of the evolution of the orbit will be determined by a first approximation of the increment $\Delta_{i1}x$. Remaining within the limits of the first approximation, we may discover a few regularities which will be convenient for understanding the process of evolution.

The change of height of perigee of the orbit r_π and the change of eccentricity with $a = \text{const}$ are connected by the simple relation $\Delta r_\pi = -a\Delta e$, obtained from the formula $r_\pi = a(1 - e)$ with a constant. The sign of the change of eccentricity is determined, according to (17), by the sign of the magnitude $\beta_3 = \xi_1 \xi_2 \Delta_k^3$. Hence it follows that, for orbits for which a first approximation is sufficient to determine a result, the ensuing regularity is true.

The height of perigee will be increased in the case where the true anomaly θ_ξ of the projection of the vector of the perturbing acceleration is found in the first or third quadrants. In the case where the true anomaly θ_ξ is found in the second or fourth quadrants, the height of perigee will be decreased.

3. Instead of the position of the node Ω and of the angular distance of the perigee from the node ω , we may consider the longitude α and the latitude φ of Laplace's vector (the vector directed from the central body to the pericentre). In the general case we call the angular distance of Laplace's vector from the plane XY the latitude φ and we also denote the angular distance between the X axis and the projection of Laplace's vector on the plane as the longitude α . The connection between ω , i , Ω , α , φ is determined by the formulae

$$\left. \begin{aligned} \sin \varphi &= \sin \omega \sin i, \\ \cos(\alpha - \Omega) &= \frac{\cos \omega}{\cos \varphi}. \end{aligned} \right\} \quad (22)$$

According to (22)

$$\cos \varphi \Delta \varphi = \cos \omega \sin i \Delta \omega + \sin \omega \cos i \Delta i,$$

$$\sin \omega \Delta \omega = \sin (\alpha - \Omega) \cos \varphi (\Delta \alpha - \Delta \Omega) + \cos (\alpha - \Omega) \sin \varphi \Delta \varphi.$$

Substituting in these formulae the increments $\Delta_{11}x$ from (17), after transformation we shall obtain formulae for the increments $\Delta_{11}\varphi$ and $\Delta_{11}\alpha$ which we may use instead of $\Delta_{11}\Omega$ and $\Delta_{11}\omega$ in system (17).

$$\left. \begin{aligned} \Delta_{11}\varphi &= 3\pi \frac{\mu_k}{\mu} \left(\frac{a}{p_k}\right)^3 \frac{\varepsilon^{\frac{1}{2}}}{\cos \varphi} [(4\beta_1 - \beta_2 - \beta_6) \cos \omega \sin i - \beta_4 \cos i], \\ \Delta_{11}\alpha &= 3\pi \frac{\mu_k}{\mu} \left(\frac{a}{p_k}\right)^3 \frac{\varepsilon^{\frac{1}{2}}}{\cos^2 \varphi} [(4\beta_1 - \beta_2 - \beta_6) \cos i + \beta_4 \cos \omega \sin i]. \end{aligned} \right\} \quad (23)$$

According to (23) for very elongated orbits with $e \rightarrow 1$ (or such that $\varepsilon \rightarrow 0$) a stability of the angular co-ordinates in absolute space is observed. For $\varepsilon \rightarrow 0$ the evolution of the orbit has the following character, the plane of the orbit quickly turns round the line of the apsides (i.e. the parameters i , Ω , ω are quickly changed), the line of the apsides itself remaining practically motionless.

5. THE CALCULATION OF THE PARAMETERS AND THEIR DERIVATIVES FOR THE CASE OF MOTION OF THE PERTURBING BODY IN AN ELLIPSE

For shortening the writing of the calculation of the values of α_i , β_i , γ_i and their derivatives we shall use below the rules for multiplying matrices, i.e. if two matrices A and B are given by the tables

$$A = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & \dots & \dots & \dots \\ a_{i1} & a_{i2} & \dots & a_{im} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{vmatrix}, \quad B = \begin{vmatrix} b_{11} & b_{12} & \dots & b_{1j} & \dots & b_{1k} \\ b_{21} & b_{22} & \dots & b_{2j} & \dots & b_{2k} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \\ b_{m1} & b_{m2} & \dots & b_{mj} & \dots & b_{mk} \end{vmatrix},$$

then their product $C = AB$ will be a matrix with elements C_{ij} with n lines and k columns; the elements C_{ij} being calculated as the sum of the twin multiplications of the elements of the i -th line of Matrix A and the j -th column of matrix B

$$C_{ij} = a_{11}b_{1j} + a_{12}b_{2j} + \dots + a_{im}b_{mj}.$$

In particular, if the matrix B contains only one column, we have the case of the multiplication of a matrix by a vector, as the result of which we get a new vector (according to the same rule).

Obviously in the majority of problems of the approximate investigation of the evolution of satellite orbits, we may consider that the motion of the perturbing body around the central is described by the formulae of the two bodies problem. The orbit of the perturbing body may be determined by parameters p_k , e_k , i_k , Ω_k , ω_k , and the position of the perturbing body in its orbit is given by the argument of its latitude u_k at moment t_0 . In this case for the quantity p_k in formulae (17)–(19) we use the parameter of the orbit of the perturbing body, and Δ_k in formulae (10), in accordance with Kepler's law of motion, will be given by $\Delta_k = 1 + e_k \cos (u_k - \omega_k)$.

Let $\mathbf{r}_k^0$ be a vector which determines the direction of the perturbing body. In the matrix writing the projections of the vector $\mathbf{r}_k^0$ on the orbital axes of co-ordinates are determined by the formula

$$\xi = D_1 U_1, \quad (24)$$

where the vectors ξ and U_1 have the form

$$\xi = \begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix}, \quad U_1 = \begin{pmatrix} \cos u_k \\ \sin u_k \end{pmatrix}. \quad (25)$$

The matrix

$$D_1 = \begin{pmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \\ d_{31} & d_{32} \end{pmatrix} \quad (26)$$

may be given in the form of the product of the three matrices (27)–(29) $D_1 = CBA$, where the matrix A determines the projections of the vector $\mathbf{r}_k^0$ on the axes of the system of co-ordinates XYZ ; matrix B determines the transition from the system XYZ to an Eulerian system of co-ordinates (one of the axes directed towards the ascending node of the satellite's orbit, the second axis orthogonal to the first and lying in the plane of the orbit of the satellite, and the third orthogonal to the plane of the satellite's orbit); matrix C determines the transition from the Eulerian system of co-ordinates to the orbital system.

$$A = \begin{pmatrix} \cos \Omega_k & -\sin \Omega_k \cos i_k \\ \sin \Omega_k & \cos \Omega_k \cos i_k \\ 0 & \sin i_k \end{pmatrix}, \quad (27)$$

$$B = \begin{pmatrix} \cos \Omega & \sin \Omega & 0 \\ -\sin \Omega \cos i & \cos \Omega \cos i & \sin i \\ \sin \Omega \sin i & -\cos \Omega \sin i & \cos i \end{pmatrix}, \quad (28)$$

$$C = \begin{pmatrix} \cos \omega & \sin \omega & 0 \\ -\sin \omega & \cos \omega & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (29)$$

The presentation of the vector ξ in the form (24) is convenient for the differentiation and integration of this vector with respect to the time or with respect to the parameter u_k . From the integral of the squares for the perturbing body we have

$$\left. \begin{aligned} \frac{du_k}{dt} &= b \Delta_k^2, \\ b &= \frac{2\pi}{T_k \varepsilon_k^{\frac{1}{2}}}, \quad \varepsilon_k = 1 - e_k^2, \end{aligned} \right\} \quad (30)$$

where

T_k is the period of revolution of the perturbing body around the central. The derivatives of higher orders of u_k with respect to the time may be obtained by the following differentiation:

$$\frac{d^2 u_k}{dt^2} = -2b^2 e_k \Delta_k^3 \sin(u_k - \omega_k) \quad (31)$$

etc.

From (24) and (30) we have

$$\frac{d\xi}{dt} = \begin{pmatrix} \frac{d\xi_1}{dt} \\ \frac{d\xi_2}{dt} \\ \frac{d\xi_3}{dt} \end{pmatrix} = D_1 \frac{dU_1}{dt} = b\Delta_k^2 D_1 \frac{dU_1}{du_k} = b\Delta_k^2 D_1 \begin{pmatrix} -\sin u_k \\ \cos u_k \end{pmatrix}. \quad (32)$$

Similarly, using (31) we have

$$\frac{d^2 \xi}{dt^2} = bD_1 \frac{d}{dt} \left(\Delta_k^2 \frac{dU_1}{du_k} \right) = b^2 \Delta_k^2 D_1 \frac{d}{du_k} \left(\Delta_k^2 \frac{dU_1}{du_k} \right) \quad (33)$$

etc.

Formulae (10), (24), (30), (32) permit us to calculate all the values of α_i , β_i , γ_i and $\frac{d\beta_i}{dt}$, entering into the system of equations (17)–(19). With this, to calculate the increments it is necessary in (24), (30), and (32) to substitute the value $u_k = u_k^*$ corresponding to the middle of the time of revolution of the satellite ($t = t^*$).

From now on it will be more convenient to use the following idea. We introduce into the consideration the vectors

$$\alpha = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \end{pmatrix}, \quad \gamma = \begin{pmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \\ \gamma_4 \\ \gamma_5 \\ \gamma_6 \\ \gamma_7 \end{pmatrix} \quad (34)$$

and the vectors

$$U_2 = \begin{pmatrix} \cos^2 u_k \\ \cos u_k \sin u_k \\ \sin^2 u_k \end{pmatrix}, \quad U_3 = \begin{pmatrix} \cos^3 u_k \\ \cos^2 u_k \sin u_k \\ \cos u_k \sin^2 u_k \\ \sin^3 u_k \end{pmatrix}. \quad (35)$$

The vectors α , β , γ according to the previous notion may be presented in the following form

$$\begin{aligned}\alpha &= D_1 U_1 \Delta_k^4, \\ \beta &= D_2 U_2 \Delta_k^3, \quad \beta_6 = \Delta_k^3 \\ \gamma &= D_3 U_3 \Delta_k^4,\end{aligned}\tag{36}$$

where D_2 and D_3 are matrices, obtained from the elements of the matrix D_1 in the following way

$$D_2 = \begin{vmatrix} d_{11}^2 & 2d_{11}d_{12} & d_{12}^2 \\ d_{21}^2 & 2d_{21}d_{22} & d_{22}^2 \\ d_{11}d_{21} & d_{11}d_{22} + d_{12}d_{21} & d_{12}d_{22} \\ d_{31}d_{21} & d_{31}d_{22} + d_{32}d_{21} & d_{32}d_{22} \\ d_{11}d_{31} & d_{12}d_{31} + d_{11}d_{32} & d_{12}d_{32} \end{vmatrix}\tag{37}$$

$$D_3 = \begin{vmatrix} d_{11}^3 & 3d_{11}^2d_{12} & 3d_{11}d_{12}^2 & d_{12}^3 \\ d_{21}^3 & 3d_{21}^2d_{22} & 3d_{21}d_{22}^2 & d_{22}^3 \\ d_{11}^2d_{21} & d_{22}d_{11}^2 + 2d_{11}d_{21}d_{12} & d_{21}d_{12}^2 + 2d_{11}d_{12}d_{22} & d_{12}^2d_{22} \\ d_{11}^2d_{31} & d_{32}d_{11}^2 + 2d_{11}d_{31}d_{12} & d_{31}d_{12}^2 + 2d_{11}d_{12}d_{32} & d_{12}^2d_{32} \\ d_{21}^2d_{31} & d_{32}d_{21}^2 + 2d_{21}d_{31}d_{22} & d_{31}d_{22}^2 + 2d_{21}d_{22}d_{32} & d_{22}^2d_{32} \\ d_{21}^2d_{11} & d_{12}d_{21}^2 + 2d_{11}d_{21}d_{22} & d_{11}d_{22}^2 + 2d_{12}d_{21}d_{22} & d_{12}d_{22}^2 \\ d_{11}d_{21}d_{31} & d_{11}d_{21}d_{32} + d_{11}d_{22}d_{31} + d_{12}d_{21}d_{31} & d_{11}d_{22}d_{32} + d_{12}d_{21}d_{32} + d_{12}d_{22}d_{31} & d_{12}d_{22}d_{32} \end{vmatrix}\tag{38}$$

The quantities $\frac{d\beta_i}{dt}$ entering into the equation (18), in accordance with (36) are calculated by the formulae

$$\left. \begin{aligned} \frac{d\beta_1}{dt} \\ \vdots \\ \frac{d\beta_5}{dt} \end{aligned} \right\} = D_2 \frac{d}{dt} (U_2 \Delta_k^3) = D_2 b \Delta_k^2 \frac{d}{du_k} (U_2 \Delta_k^3)\tag{39}$$

$$\frac{d\beta_6}{dt} = -3e_k b \Delta_k^4 \sin(u_k - \omega_k).$$

The components of the vector $\frac{d}{du_k} (U_2 \Delta_k^3)$ we obtain by formal differentiation in the following way

$$\frac{d}{du_k} (U_2 \Delta_k^3) = \begin{vmatrix} -2\Delta_k^3 \cos u_k \sin u_k - 3e_k \Delta_k^2 \sin(u_k - \omega_k) \cos^2 u_k \\ \Delta_k^3 \cos 2u_k - 3e_k \Delta_k^2 \sin(u_k - \omega_k) \cos u_k \sin u_k \\ 2\Delta_k^3 \cos u_k \sin u_k - 3e_k \Delta_k^2 \sin(u_k - \omega_k) \sin^2 u_k \end{vmatrix}.$$

6. FORMULAE FOR THE CHANGE OF ELEMENTS OF AN ORBIT FOR A SMALL NUMBER OF REVOLUTIONS OF THE SATELLITE

In the case of the approximate discussion it may also be convenient, for the defined class of orbit sufficiently near to the central body, in calculating the increments of the elements of the satellite's orbit, to neglect on the right-hand side of the equations the change in these elements, not only for one revolution (as was assumed in Section 3) but also for a series of subsequent revolutions of the satellite. For this it is shown to be impossible to determine the total increment of the element by simple multiplication of the increment, for example, during the first of the revolutions considered, by the number of revolutions of the satellite, since even with fixed orbital elements the relations (17)–(19) retain the parameters α_i , β_i , γ_i which, generally speaking, change appreciably with the movement of the perturbing body.

Let $\Delta_m x$ be the increment of one of the elements of the orbit for the m -th revolution. According to the approximate formulae, $\Delta_m x$ with invariable parameters of orbit is a function of the middle moment of time for the given revolution t_m^* or else $\Delta_m x$ is a function of u_m^* . The increment δx for N subsequent revolutions will be the sum

$$\delta x = \sum_{m=1}^N \Delta_m x$$

Let $\Delta t = T$ the period of the satellite's revolution. Then δx may be written down in the form of an integral sum

$$\delta x = \frac{1}{T} \sum_{m=1}^N \Delta_m x \cdot \Delta t. \quad (40)$$

We shall now assume that the case is being considered, where the relation between the period of revolution of the satellite T and the period of revolution of the perturbing body T_k , on which on an average the speed of change of the position of the vector $\mathbf{r}_k^0$ depends, is such that the sum (40) may be replaced by the integral

$$\delta x = \frac{1}{T} \int_{t_0}^{t_N} \Delta x \, dt, \quad (41)$$

where t_0 is the initial moment, and $t_N - t_0$ is the time, in the course of which the satellite will cover N revolutions.

Assuming that the elements $p, e, \dots, \omega$ on the right hand sides of equations (17)–(19) are invariable during the time of these N revolutions of the satellite, because of the linearity of formulae (17)–(19) in $\alpha_i, \beta_i, \gamma_i, \frac{d\beta_i}{dt}$ the corresponding formulae for increments $\delta_{11}x, \delta_{12}x, \delta_{21}x, \dots$ for N revolutions are obtained by the simple replacement of the values $\alpha_i, \beta_i, \gamma_i, \frac{d\beta_i}{dt}$ in formulae (17)–(19) by the integrals

$$\frac{1}{T} \int_{t_0}^{t_N} \alpha_i \, dt, \frac{1}{T} \int_{t_0}^{t_N} \beta_i \, dt, \frac{1}{T} \int_{t_0}^{t_N} \gamma_i \, dt, \frac{1}{T} \int_{t_0}^{t_N} \left(\frac{d\beta_i}{dt} \right) dt.$$

We shall call the integrals

$$\frac{1}{T} \int_{t_0}^{t_N} \alpha_i dt = [\alpha_i], \quad \frac{1}{T} \int_{t_0}^{t_N} \beta_i dt = [\beta_i], \quad \frac{1}{T} \int_{t_0}^{t_N} \gamma_i dt = [\gamma_i],$$

$$\frac{1}{T} \int_{t_0}^{t_N} \left(\frac{d\beta_i}{dt} \right) dt = \left[\frac{d\beta_i}{dt} \right].$$

For the case of motion of the perturbing body around the central, by Kepler's laws of motion, according to (30), (36), and (39) the following relations are true:

$$\left. \begin{aligned} [\alpha] &= \begin{bmatrix} [\alpha_1] \\ [\alpha_2] \\ [\alpha_3] \end{bmatrix} = D_1 \mathbf{V}_1, & [\beta] &= \begin{bmatrix} [\beta_1] \\ [\beta_2] \\ \cdot \\ \cdot \\ [\beta_5] \end{bmatrix} = D_2 \mathbf{V}_2, \\ \\ [\gamma] &= \begin{bmatrix} [\gamma_1] \\ [\gamma_2] \\ \cdot \\ \cdot \\ [\gamma_7] \end{bmatrix} = D_3 \mathbf{V}_3, & \left[\frac{d\beta}{dt} \right] &= \begin{bmatrix} \frac{d\beta_1}{dt} \\ \frac{d\beta_2}{dt} \\ \cdot \\ \cdot \\ \frac{d\beta_5}{dt} \end{bmatrix} = D_4 \mathbf{V}_4, \end{aligned} \right\} \quad (42)$$

$$\left. \begin{aligned} \mathbf{V}_1 &= \frac{1}{Tb} \int_{u_0}^{u_N} \mathbf{U}_1 \Delta_k^2 du_k, \\ \mathbf{V}_2 &= \frac{1}{Tb} \int_{u_0}^{u_N} \mathbf{U}_2 \Delta_k du_k, \\ \mathbf{V}_3 &= \frac{1}{Tb} \int_{u_0}^{u_N} \mathbf{U}_3 \Delta_k^2 du_k, \\ \mathbf{V}_4 &= \frac{1}{T} \int_{t_0}^{t_N} \frac{d}{dt} (\mathbf{U}_2 \Delta_k^3) dt = \frac{1}{T} \{ \mathbf{U}_2 \Delta_k^3 \}_{u_0}^{u_N} \end{aligned} \right\} \quad (43)$$

where u_0 and u_N are the values of the argument of the latitude of the perturbing body at the moments of time t_0 and t_N . In the formulae for the vector $\mathbf{V}_4$ the usual notations are

employed for the difference between the values of the function at two points $\{f(u)\}_{u_0}^{u_N} = f(u_N) - f(u_0)$.

The quantity $[\beta_\delta]$, not included in formulae (42) is calculated directly

$$[\beta_\delta] = \frac{1}{T} \int_{t_0}^{t_N} \Delta_k^3 dt = \frac{1}{Tb} \int_{u_0}^{u_N} \Delta_k du_k = \frac{1}{Tb} \{u_k + e_k \sin(u_k - \omega_k)\}_{u_0}^{u_N}. \quad (44)$$

After the calculation of the integrals in (43) for the components V_{ij} of the vectors $\mathbf{V}$ we obtain formulae (index i is associated with the vector and the index j with the respective number of the component)

$$\left. \begin{aligned} V_{11} &= \frac{1}{bT} \left\{ \sin u_k + e_k \left[\cos \omega_k \left(u_k + \frac{1}{2} \sin 2u_k \right) + \sin \omega_k \sin^2 u_k \right] \right. \\ &\quad \left. + e_k^2 \left[\cos^2 \omega_k \left(\sin u_k - \frac{1}{3} \sin^3 u_k \right) - \frac{2}{3} \sin \omega_k \cos \omega_k \cos^3 u_k \right. \right. \\ &\quad \left. \left. + \frac{1}{3} \sin^2 \omega_k \sin^3 u_k \right] \right\}_{u_0}^{u_N}, \\ V_{12} &= \frac{1}{bT} \left\{ -\cos u_k + e_k \left[\cos \omega_k \sin^2 u_k + \sin \omega_k \left(u_k - \frac{1}{2} \sin 2u_k \right) \right] \right. \\ &\quad \left. + e_k^2 \left[-\frac{1}{3} \cos^2 \omega_k \cos^3 u_k + \frac{2}{3} \sin \omega_k \cos \omega_k \sin^3 u_k \right. \right. \\ &\quad \left. \left. - \sin^2 \omega_k \left(\cos u_k - \frac{1}{3} \cos^3 u_k \right) \right] \right\}_{u_0}^{u_N}, \end{aligned} \right\} \quad (45)$$

$$\left. \begin{aligned} V_{21} &= \frac{1}{bT} \left\{ \frac{1}{2} u_k + \frac{1}{2} \sin u_k \cos u_k + e_k \left[\cos \omega_k \left(\sin u_k - \frac{1}{3} \sin^3 u_k \right) \right. \right. \\ &\quad \left. \left. - \frac{1}{3} \sin \omega_k \cos^2 u_k \right] \right\}_{u_0}^{u_N}, \\ V_{22} &= \frac{1}{bT} \left\{ \frac{1}{2} \sin^2 u_k + \frac{1}{3} e_k (-\cos \omega_k \cos^3 u_k + \sin \omega_k \sin^3 u_k) \right\}_{u_0}^{u_N}, \\ V_{23} &= \frac{1}{bT} \left\{ \frac{1}{2} u_k - \frac{1}{2} \sin u_k \cos u_k + e_k \left[\frac{1}{3} \cos \omega_k \sin^3 u_k \right. \right. \\ &\quad \left. \left. + \sin \omega_k \left(-\cos u_k + \frac{1}{3} \cos^3 u_k \right) \right] \right\}_{u_0}^{u_N}, \end{aligned} \right\} \quad (46)$$

$$\begin{aligned}
 V_{31} &= \frac{1}{bT} \left\{ \sin u_k - \frac{1}{3} \sin^3 u_k + e_k \left[\cos \omega_k \left(\frac{3}{4} u_k + \sin u_k \cos u_k \right. \right. \right. \\
 &\quad \left. \left. + \frac{1}{4} \sin u_k \cos u_k \cos 2u_k \right) - \frac{1}{2} \sin \omega_k \cos^4 u_k \right] \\
 &\quad \left. + e_k^2 \left[\cos^2 \omega_k \left(\sin u_k - \frac{2}{3} \sin^3 u_k + \frac{1}{5} \sin^5 u_k \right) \right. \right. \\
 &\quad \left. \left. - \frac{2}{5} \cos \omega_k \sin \omega_k \cos^5 u_k + \sin^2 \omega_k \left(\frac{1}{3} \sin^3 u_k - \frac{1}{5} \sin^5 u_k \right) \right] \right\}_{u_0}^{u_N}, \\
 V_{32} &= \frac{1}{bT} \left\{ -\frac{1}{3} \cos^3 u_k + e_k \left[-\frac{1}{2} \cos \omega_k \cos^4 u_k \right. \right. \\
 &\quad \left. \left. + \sin \omega_k \left(\frac{1}{4} u_k - \frac{1}{4} \sin u_k \cos u_k \cos 2u_k \right) \right] \right. \\
 &\quad \left. + e_k^2 \left[-\frac{1}{5} \cos^2 \omega_k \cos^5 u_k + 2 \cos \omega_k \sin \omega_k \left(\frac{1}{3} \sin^3 u_k \right. \right. \right. \\
 &\quad \left. \left. - \frac{1}{5} \sin^5 u_k \right) - \sin^2 \omega_k \left(\frac{1}{3} \cos^3 u_k - \frac{1}{5} \cos^5 u_k \right) \right] \right\}_{u_0}^{u_N}, \\
 V_{33} &= \frac{1}{bT} \left\{ \frac{1}{3} \sin^3 u_k + e_k \left[\cos \omega_k \left(\frac{1}{4} u_k - \frac{1}{4} \sin u_k \cos u_k \cos 2u_k \right) \right. \right. \\
 &\quad \left. \left. + \frac{1}{2} \sin \omega_k \sin^4 u_k \right] + e_k^2 \left[\cos^2 \omega_k \left(\frac{1}{3} \sin^3 u_k - \frac{1}{5} \sin^5 u_k \right) \right. \right. \\
 &\quad \left. \left. - 2 \sin \omega_k \cos \omega_k \left(\frac{1}{3} \cos^3 u_k - \frac{1}{5} \cos^5 u_k \right) + \frac{1}{5} \sin^2 \omega_k \sin^5 u_k \right] \right\}_{u_0}^{u_N}, \\
 V_{34} &= \frac{1}{bT} \left\{ -\left(\cos u_k - \frac{1}{3} \cos^3 u_k \right) + e_k \left[\frac{1}{2} \cos \omega_k \sin^4 u_k \right. \right. \\
 &\quad \left. \left. + \sin \omega_k \left(\frac{3}{4} u_k - \sin u_k \cos u_k + \frac{1}{4} \sin u_k \cos u_k \cos 2u_k \right) \right] \right. \\
 &\quad \left. + e_k^2 \left[-\cos^2 \omega_k \left(\frac{1}{3} \cos^3 u_k - \frac{1}{5} \cos^5 u_k \right) + \frac{2}{5} \sin \omega_k \cos \omega_k \sin^5 u_k \right. \right. \\
 &\quad \left. \left. - \sin^2 \omega_k \left(\cos u_k - \frac{2}{3} \cos^3 u_k + \frac{1}{5} \cos^5 u_k \right) \right] \right\}_{u_0}^{u_N}.
 \end{aligned} \tag{47}$$

7. THE INVESTIGATIONS OF THE EQUATIONS OF SECULAR CHANGES IN THE ELEMENTS FOR A PERIOD OF REVOLUTION OF THE PERTURBING BODY.

In the case of orbits of satellites, for which the assumption of Section 6, concerning the fixed values of the elements of the orbit for a few revolutions with a sufficient degree of approximation, is permissible for every period of revolution of the perturbing body T_k , we may obtain particularly simple formulae for the average change of the elements throughout a revolution of the perturbing body.

We shall substitute in formulae (45)–(47) $u_N = u_0 + 2\pi$. As a result we shall get

$$\left. \begin{aligned} V_{11} &= \frac{2\pi e_k \cos \omega_k}{bT}, \\ V_{12} &= \frac{2\pi e_k \sin \omega_k}{bT}, \end{aligned} \right\} \quad (48)$$

$$\left. \begin{aligned} V_{21} &= \frac{\pi}{bT}, \\ V_{22} &= 0, \\ V_{23} &= \frac{\pi}{bT}, \end{aligned} \right\} \quad (49)$$

$$\left. \begin{aligned} V_{31} &= \frac{3\pi e_k}{2bT} \cos \omega_k, \\ V_{32} &= \frac{1\pi e_k}{2bT} \sin \omega_k, \\ V_{33} &= \frac{1\pi e_k}{2bT} \cos \omega_k, \\ V_{34} &= \frac{3\pi e_k}{2bT} \sin \omega_k. \end{aligned} \right\} \quad (50)$$

The vector $\mathbf{V}_4 = 0$ according to (43). It is easily seen that, in the case considered, all increments $\delta_{ij}x$ for a revolution of the perturbing body for any i and for $j > 1$ are zero, since under the integrals (41) will stand the total differentials, with respect to the time, of periodic functions with the period of the perturbing body.

The increments $\delta_{21}x$ are proportional to the magnitudes of the components of the vectors $\mathbf{V}_1$ and $\mathbf{V}_3$, which in their turn, according to (48) and (50), are proportional to the eccentricity of the orbit of the perturbing body e_k . In accordance with this the ratio of the increments

$\delta_{21}x$ to $\delta_{11}x$ has the order of $\frac{a}{p_k} e_k$.

If we investigate the secular departures for a period of revolution of the perturbing body, having an orbit with small eccentricity then, if our basic assumption is correct, $\frac{a}{p_k} \ll 1$ and (with the fulfilment of all the other prerequisites) we may often neglect the increments $\delta_{21}x$.

In the present section we shall limit ourselves to the investigation of the increments $\delta_{11}x$.

In accordance with the determination of the quantity b of (30) we have $\frac{\pi}{bT} = \frac{1}{2} \frac{T_k}{T} e_k^{\frac{3}{2}}$.

Here $\frac{T_k}{T} \approx N$ —the number of revolutions of the satellite in a period of revolution of the perturbing body.

If for the calculation of $\delta_{11}x$ we avail ourselves of the vector $\frac{\mathbf{V}_2}{N}$ we shall obtain the formulae

for the changes in the elements of the satellite's orbit for one revolution, averaged over the period of the perturbing body.

We shall take as the basic system of measurement an XYZ system of co-ordinates whose plane XY coincides with the orbit of the perturbing body. In this case $i_k = 0$. The parameter Ω_k may also be assumed to be zero, considering that ω_k is measured from the axis of X .

From physical considerations it is clear that the average increments $\delta_{11}x$ generally speaking, must depend on the angle $\Omega - \omega_k$ between the node of the satellite's orbit and the perigee of the orbit of the perturbing body, and not on each of these parameters separately. According to (26)–(29) the matrix D_1 (and consequently also the matrices D_2 and D_3) does

not depend on ω_k . According to (49) the vector $\frac{\mathbf{V}_2}{N}$ also does not depend on ω_k . Therefore the increment $\delta_{11}x$ will not depend on ω_k and consequently also on the position of the node of the orbit of the satellite Ω . Thus it is possible, with the choice of average ratios for the increments, to assume that $\Omega = 0$.

Assuming in (27)–(29) $i_k = \Omega_k = \Omega = 0$, we have

$$D_1 = CBA = \begin{vmatrix} \cos \omega & \cos i \sin \omega \\ -\sin \omega & \cos i \cos \omega \\ 0 & -\sin i \end{vmatrix}. \quad (51)$$

In this case the matrix D_2 , according to (37) has the form

$$D_2 = \begin{vmatrix} \cos^2 \omega & x_1 & \cos^2 i \sin^2 \omega \\ \sin^2 \omega & x_2 & \cos^2 i \cos^2 \omega \\ -\sin \omega \cos \omega & x_3 & \cos^2 i \sin \omega \cos \omega \\ 0 & x_4 & -\sin i \cos i \cos \omega \\ 0 & x_5 & -\sin i \cos i \sin \omega \end{vmatrix} \quad (52)$$

It is not necessary to calculate the second column of the matrix D_2 since the component $V_{22} = 0$.

We obtain the average values of the quantity $\bar{\beta}_i$ with the help of (42), using the matrix D_2 in the form (52) and the components of the vector $\frac{\mathbf{V}_2}{N}$, determined in accordance with (49).

In addition we get $\bar{\beta}_i$ from (44) with $u_N = u_o + 2\pi$:

$$\left. \begin{aligned} \bar{\beta}_1 &= \frac{1}{2} \varepsilon_k^{\frac{3}{2}} (\cos^2 \omega + \cos^2 i \sin^2 \omega), \\ \bar{\beta}_2 &= \frac{1}{2} \varepsilon_k^{\frac{3}{2}} (\sin^2 \omega + \cos^2 i \cos^2 \omega), \\ \bar{\beta}_3 &= -\frac{1}{2} \varepsilon_k^{\frac{3}{2}} \sin^2 i \sin \omega \cos \omega, \\ \bar{\beta}_4 &= -\frac{1}{2} \varepsilon_k^{\frac{3}{2}} \sin i \cos i \cos \omega, \\ \bar{\beta}_5 &= -\frac{1}{2} \varepsilon_k^{\frac{3}{2}} \sin i \cos i \sin \omega, \\ \bar{\beta}_6 &= \varepsilon_k^{\frac{3}{2}}. \end{aligned} \right\} \quad (53)$$

Substituting $\tilde{\beta}_i$ from (53) in (17) instead of the corresponding β_i we shall obtain equations for the changes of the elements of the orbit in one revolution, averaged over a period of revolution of the perturbing body. These changes we shall denote by $\frac{\delta x}{\delta N}$:

$$\left. \begin{aligned} \frac{\delta a}{\delta N} &= 0, \\ \frac{\delta e}{\delta N} &= \frac{1}{2} A e \varepsilon^{\frac{1}{2}} \sin^2 i \sin 2\omega, \\ \frac{\delta i}{\delta N} &= -\frac{1}{2} A \frac{(1-\varepsilon)}{\varepsilon^{\frac{1}{2}}} \sin i \cos i \sin 2\omega, \\ \frac{\delta \Omega}{\delta N} &= -A \frac{\cos i}{\varepsilon^{\frac{1}{2}}} \left[(1-\varepsilon) \sin^2 \omega + \frac{1}{5} \varepsilon \right], \\ \frac{\delta \omega}{\delta N} &= A \frac{1}{\varepsilon^{\frac{1}{2}}} \left[(\cos^2 i - \varepsilon) \sin^2 \omega + \frac{2}{5} \varepsilon \right], \end{aligned} \right\} \quad (54)$$

$$A = \frac{15}{2} \pi \frac{\mu_k}{\mu} \left(\frac{a}{p_k} \right)^3 \varepsilon_k^{\frac{3}{2}}. \quad (55)$$

For analysis it is sometimes more convenient to replace the last two equations of (54) by the equations for the longitude α and latitude φ of the line of apsides. The equation for α and φ may be obtained either from (54) using the relations (22), or by direct substitution of the quantities $\tilde{\beta}_i$ in (23)

$$\left. \begin{aligned} \frac{\delta \varphi}{\delta N} &= A \varepsilon^{\frac{1}{2}} \frac{\sin i \cos \omega}{\cos \varphi} \left[\frac{2}{5} - \sin^2 \varphi \right], \\ \frac{\delta \alpha}{\delta N} &= \frac{1}{5} A \varepsilon^{\frac{1}{2}} \frac{\cos i}{\cos^2 \varphi} [1 - 4 \sin^2 \varphi]. \end{aligned} \right\} \quad (56)$$

Instead of the equation for the change of eccentricity (the second of the equations (54)) we may consider the equivalent equation for ε

$$\frac{\delta \varepsilon}{\delta N} = -A(1-\varepsilon) \varepsilon^{\frac{1}{2}} \sin^2 i \sin 2\omega. \quad (57)$$

From the first two of equations (54) and the relation $r_\pi = a(1-e)$ there results the following regularity in the relation of the sign of the secular departure of the height of the pericentre during a revolution of the perturbing body: the height of the pericentre will be increased in the case where the angular distance ω of the pericentre from the ascending node to the plane of the orbit of the perturbing body belongs to the second or fourth quadrant and accordingly a decrease of height of the pericentre will take place if ω belongs to the first or third quadrants.

Below an investigation of system (54) is carried out. For this assuming that the increments δx are sufficiently small, we shall consider, by a well-known method, the ratios of the variations $\frac{\delta x}{\delta N}$ as ordinary derivatives.

The system (54), except for the trivial integral $a = \text{const}$, in the general case possess two integrals:

$$\varepsilon = \frac{c_1}{\cos^2 i} \quad (58)$$

and

$$1 - \varepsilon = \frac{c_2}{\frac{2}{5} - \sin^2 i \sin^2 \omega} \quad (59)$$

where c_1 and c_2 are constants, determined by the initial values of ε_0 , i_0 , ω_0 .

$$c_1 = \varepsilon_0 \cos^2 i_0, \quad c_2 = (1 - \varepsilon_0) \left(\frac{2}{5} - \sin^2 i_0 \sin^2 \omega_0 \right) \quad (60)$$

Integrals (58) and (59) determine the dependence $\varepsilon = f_1(\omega)$ and $\cos i = f_2(\omega)$, and the solution of every problem is found by two quadratures

$$N - N_0 = \frac{1}{A} \int_{\omega_0}^{\omega} \frac{\varepsilon^{\frac{1}{2}} d\omega}{\left[(\cos^2 i - \varepsilon) \sin^2 \omega + \frac{2}{5} \varepsilon \right]} \quad (61)$$

$$\Omega - \Omega_0 = -A \int_{N_0}^N \frac{\cos i}{\varepsilon^{\frac{1}{2}}} \left[(1 - \varepsilon) \sin^2 \omega + \frac{1}{5} \varepsilon \right] dN. \quad (62)$$

Special cases of solutions of system (54).

1. Let $\sin i_0 = 0$, i.e. $i_0 = 0$ or $i_0 = \pi$.

From equations (54) and (56) it follows that

$$\left. \begin{aligned} \varepsilon = \varepsilon_0 = \text{const}, \quad \frac{\delta \alpha}{\delta N} = \pm \frac{1}{5} A \varepsilon_0^{\frac{1}{2}} \\ i = 0, \quad \varphi = 0, \end{aligned} \right\} \quad (63)$$

(the plus sign corresponds to $i_0 = 0$ and the minus sign to $i_0 = \pi$) i.e. in this case, if the orbit of the satellite at the initial moment lies in the plane of the orbit of the perturbing body, all parameters (with the exception of the longitude of the line of apsides) remain on an average unchanged. The line of apsides turns in the plane of the orbit of the perturbing body on an average with constant speed.

2. Let $\cos i_0 = 0$.

In this case $i = i_0 = \frac{\pi}{2}$, $\Omega = \Omega_0 = \text{const}$. The parameters ε and ω are connected by the integral (59)

$$1 - \varepsilon = \frac{c_2''}{5 \cos 2\omega - 1}, \quad (64)$$

$$c_2'' = (1 - \varepsilon_0)(5 \cos 2\omega_0 - 1). \quad (65)$$

Here it is assumed that $\omega_0 \neq \omega_1^*$, where $\omega_1^* = \frac{1}{2} \arccos \frac{1}{5}$; $\omega_2^* = -\omega_1^*$; $\omega_3^* = \pi - \omega_1^*$; $\omega_4^* = \pi + \omega_1^*$.

From the last of equations (54) and in accordance with (64) we get an equation for $\omega(N)$

$$\frac{\delta \omega}{\delta N} = \frac{A}{10} (5 \cos 2\omega - 1) \sqrt{1 - \frac{c_2''}{5 \cos 2\omega - 1}}. \quad (66)$$

The character of the solution depends on the initial value of ω_0 .

(a) Let $\omega_2^* < \omega_0 < \omega_1^*$ (or $\omega_3^* < \omega_0 < \omega_4^*$). In this case $\delta\omega > 0$ and ω increases monotonically.

If $\omega_2^* < \omega_0 < 0$ (or $\omega_3 < \omega_0 < \pi$) the solution has the following character. Up to these times while $\omega < 0$ ($\omega < \pi$), $\delta\varepsilon > 0$ and the eccentricity of the orbit is decreasing, reaching with $\omega = 0$ ($\omega = \pi$) a minimum value, which may be obtained from (64),

$$e_{\min}^2 = \frac{e_0^2(5 \cos 2\omega_0 - 1)}{4}. \quad (67)$$

With further growth of ω the eccentricity of the orbit increases and with $\omega = \bar{\omega}$, where

$$5 \cos 2\bar{\omega} - 1 = (1 - \varepsilon_0)(5 \cos 2\omega_0 - 1), \quad (68)$$

ε changes to zero, (i.e. $e = 1$).

From (66) it follows that the value $\omega = \bar{\omega}$ is reached during the final number of the revolutions. The choice of quadrant for $\bar{\omega}$ is derived from the condition that $\bar{\omega}$ belongs to the same interval of ω as does ω_0 . From (68) it follows that $\bar{\omega} < \omega_1^*$ (or in the corresponding case $\bar{\omega} < \omega_4^*$).

If the central body has radius R , the fall to the surface takes place before ε changes to zero, namely for

$$\tilde{\varepsilon} = 1 - \left(1 - \frac{R}{a}\right)^2 \quad (69)$$

and $\tilde{\omega}$ is determined by (64)

$$5 \cos 2\tilde{\omega} - 1 = \frac{(1 - \varepsilon_0)(5 \cos 2\omega_0 - 1)}{1 - \tilde{\varepsilon}}. \quad (70)$$

If $0 < \omega_0 < \omega_1^*$ or $\pi < \omega_0 < \omega_4^*$ there will be no part with decreasing eccentricity. In other respects the character of the evolution of the orbit coincides with the part of the case where the eccentricity is increasing.

(b) In the case $\omega_1^* < \omega_0 < \omega_3^*$ or $\omega_4^* < \omega_0 < 2\pi + \omega_2^*$ we have $\delta\omega < 0$ and ω diminishes from its initial value ω_0 to the corresponding value $\tilde{\omega}$, determined by (69) and (70).

In this case $\omega_1^* < \tilde{\omega} < \frac{\pi}{2}$ ($\omega_4^* < \omega < \frac{3}{2}\pi$). If in the case considered $\frac{\pi}{2} < \omega_0 < \omega_3^*$ ($\frac{3}{2}\pi < \omega_0 < 2\pi + \omega_2^*$) initially there is a decrease of eccentricity to the minimum value, which according to (64) is equal to

$$e_{\min}^2 = \frac{e_0^2(1 - 5 \cos 2\omega_0)}{6}, \quad (71)$$

and following that an increase to the value determined by (69) at the moment of fall to the surface of the central body.

If ω_0 is already less than $\frac{\pi}{2}$ (or $\frac{3}{2}\pi$ respectively) the evolution begins at once with an increase of eccentricity.

(c) In the case $\omega_0 = \omega_i^*$ we have $\omega = \omega_i^* = \text{const}$ and from (57) it follows

$$\frac{\delta\varepsilon}{\delta N} = -A(1 - \varepsilon)\varepsilon^{\frac{1}{2}} \sin 2\omega_i^*. \quad (72)$$

With ω_0 equal to ω_1^* or ω_4^* , ε decreases monotonically to the value $\tilde{\varepsilon}$ determined by (69) and corresponds to a fall to the surface of the central body. In the case $\omega_0 = \omega_2^*$ or $\omega_0 = \omega_3^*$ there is a monotonic increase and also the approach of the orbit to a circle. However,

from analysis of the case (b), we may conclude that the solution for $\omega_0 = \omega_2^*$ or $\omega_0 = \omega_3^*$ is unstable.

From the foregoing discussion it follows that, with the exclusion of the special initial conditions $\omega_0 = \omega_2^*$ and $\omega_0 = \omega_3^*$, all orbits, whose planes are orthogonal to the plane of the perturbing body, evolve in such a way that the satellite falls onto the central body during the last interval of time. With a special choice of ω_0 we may obtain a temporary significant decrease of the original eccentricity of the orbit of the satellite compared with the initial. From the formulae for $e_{\min}$ it follows that a much larger decrease of e may be obtained in the case where ω_0 is sufficiently near to ω_2^* or ω_3^* .

3. Let $\sin^2 \varphi_0 = \sin^2 i_0 \sin^2 \omega_0 = \frac{2}{5}$

In this case $\varphi = \varphi_0 = \text{const}$ and there are two integrals

$$\varepsilon = \frac{\varepsilon_0 \cos^2 i_0}{\cos^2 i}, \quad \sin^2 \omega = \frac{2}{5} \frac{1}{\sin^2 i}.$$

The dependence $\omega(N)$ is determined from the equation

$$\frac{\delta \omega}{\delta N} = A \frac{(1 - \varepsilon)}{\varepsilon^{\frac{1}{2}}} \left(\sin^2 \omega - \frac{2}{5} \right). \quad (73)$$

From the second integral it follows that $\sin^2 i_0 \geq \frac{2}{5}$, $\sin^2 \omega \geq \frac{2}{5}$ and $\delta \omega \geq 0$.

The sign of equality in the last relation may have a place only for $i_0 = 90^\circ$ i.e. only in case (c) considered above (in paragraph 2).

In the present paragraph 3 we may distinguish two cases

$$(a) \arcsin \sqrt{\frac{2}{5}} \leq i_0 < \frac{\pi}{2}; \quad b) \frac{\pi}{2} < i_0 \leq \pi - \arcsin \sqrt{\frac{2}{5}}$$

In both cases a solution may exist for ω lying in the intervals (ω_1^*, ω_3^*) and $(\omega_4^*, \omega_2^* + 2\pi)$.

When $\omega < \frac{\pi}{2}$ (or $\omega = \frac{3}{2}\pi$ for the second interval) with the growth of ω the eccentricity of the orbit grows and the inclination of the orbit in case (a) decreases and in case (b) increases.

When ω reaches the value $\frac{\pi}{2}$ (or $\frac{3}{2}\pi$) the quantity ε reaches its minimum value

$$\varepsilon_{\min} = \frac{5}{3} \varepsilon_0 \cos^2 i_0$$

The inclination at this point has a value $i = \arcsin \sqrt{\frac{2}{5}}$ in case (a) and $i = \pi - \arcsin \sqrt{\frac{2}{5}}$ in case (b).

With further increase in ω the eccentricity of the orbit decreases and the orbit quickly approaches a circle, and the inclination approaches 90° and may reach a value determined by the formula

$$\cos^2 i = \varepsilon_0 \cos^2 i_0$$

However with ε sufficiently near to 1 the resulting formulae are already unfit for the description of the average evolution of the orbit.

4. With ω_0 and i_0 satisfying the condition $\cos \omega_0 = 0$ and $\cos^2 i_0 = \frac{3}{5} \varepsilon_0$, and any ε_0 as the solution in the system (54), there is an orbit on an average with constant elements $\omega = \omega_0 =$

$\pm \frac{\pi}{2}$, $\varepsilon = \varepsilon_0$, $i = i_0$, $\varphi = \varphi_0$, where $\sin^2 \varphi_0 = 1 - \frac{3}{5} \varepsilon_0$.

The longitude of the line of apsides α satisfies the equation

$$\frac{\delta\alpha}{\delta N} = \pm \frac{1}{5} A \sqrt{\frac{5}{3}} \left(\frac{12}{5} \varepsilon_0 - 3 \right), \quad (74)$$

where the plus sign is taken with $\cos i_0 > 0$, and the minus with $\cos i_0 < 0$. In this case the whole evolution on an average is shown as a rotation of the orbit around the normal to the plane of motion of the perturbing body.

The general case of evolution of an orbit with arbitrary initial data

The evolution is determined by the constants c_1 , and c_2 in the integrals (58) and (59). From these we have

$$c_1 = \varepsilon_0 \cos^2 i_0 < 1, \quad c_2 = (1 - \varepsilon_0) \left[\frac{2}{5} - \left(1 - \frac{c_1}{\varepsilon_0} \right) \sin^2 \omega_0 \right].$$

In the case where $c_2 > 0$ it is easily seen that the maximum value of c_2 for invariable ε_0 , i_0 is obtained with $\sin^2 \omega_0 = 0$ and reaches the value

$$c_{2,\max}(c_1, i_0) = \frac{2}{5} (1 - \varepsilon_0) = \frac{2}{5} \left(1 - \frac{c_1}{\cos^2 i_0} \right)$$

Moreover it follows that with fixed c_1 the greatest value of $c_{2,\max}(c_1, i_0)$ is obtained with $\cos^2 i_0 = 1$ i.e. for $i_0 = 0$ (or $i_0 = \pi$); in this case we have

$$c_{2,\max} = \frac{2}{5} (1 - c_1), \quad c_1 = \varepsilon_0.$$

In the case where $c_2 < 0$ we use the notation $c_2' = -c_2$. The maximum value of c_2' (for fixed values of ε_0 and c_1) is reached for $\sin^2 \omega_0 = 1$ and is equal to $c_{2,\max}'(c_1, \varepsilon_0) = (1 - \varepsilon_0) \left(\frac{3}{5} - \frac{c_1}{\varepsilon_0} \right)$. In its turn, considering $c_{2,\max}'(c_1, \varepsilon_0)$ as a function of ε_0 and finding its extreme value with respect to ε_0 , we find that the largest possible values of $c_{2,\max}'(c_1, \varepsilon_0)$ occur with $\varepsilon_0 = \sqrt{\frac{5}{3}} c_1$. For a given c_1 the largest possible values of c_2' (or the least values of c_2) are determined by the relation $c_{2,\max}' = -c_{2,\min} = \frac{3}{5} (1 - \sqrt{\frac{5}{3}} c_1)^2 = \frac{3}{5} (1 - \varepsilon_0)^2$.

The regions of possible values of the parameters c_1 and c_2 are given in Fig. 1. In the plane (c_1, c_2) characteristic points have co-ordinates $A(1, 0)$; $B(0, \frac{2}{5})$; $O(0, 0)$; $E(0, -\frac{2}{5})$; $D(\frac{2}{5}, 0)$.

The line AB , bounding the region of possible values of the parameters, $c_1 > 0$ and $c_2 > 0$, is determined by the equation $c_2 = \frac{2}{5} (1 - c_1)$. Following from the analysis given above, such a connection corresponds to $\cos^2 i_0 = 1$ i.e. to the special case of the solutions of system (54) investigated in paragraph 1 of the given section.

The line OA corresponds to the case $c_2 = 0$ (i.e. $\sin^2 \varphi_0 = \frac{2}{5}$). The sum total of such solutions was discussed in paragraph 3.

The line BE corresponds to $c_1 = 0$ which, according to the discussion, answers to orbits which are orthogonal to the plane of the orbit of the perturbing body; the evolution of such orbits was examined in paragraph 2.

The line ED answers to the initial values of the parameters $\sin^2 \omega_0 = 1$ and $\cos^2 i_0 = \frac{2}{5} \varepsilon_0$ for which, following from paragraph 4, the evolution bears the character of a simple turning of the orbit about the normal to the plane of the perturbing body.

As a result the boundary lines of the region of possible values of the parameters c_1 and c_2 answer to the series of special cases of initial data considered above.

From the last equation of (54) it follows that the sign of $\delta\omega$ is determined by the sign of the expression

$$(\cos^2 i - \varepsilon) \sin^2 \omega + \frac{2}{5} \varepsilon. \quad (75)$$

For $\frac{c_1}{\varepsilon^2} > 1$ with the calculation (58) we have $\delta\omega \geq 0$ for any ω . With $\frac{c_1}{\varepsilon^2} < 1$, for the satisfaction of the condition $\delta\omega \geq 0$ it also turns out from (75), (58) and (59) to be necessary to satisfy the inequality

$$\frac{\frac{2}{5} - \frac{c_2}{1 - \varepsilon}}{1 - \frac{c_1}{\varepsilon}} \leq \frac{\frac{2}{5}}{1 - \frac{c_1}{\varepsilon^2}}. \quad (76)$$

So long as in the case considered $1 - \frac{c_1}{\varepsilon} > 1 - \frac{c_1}{\varepsilon^2} > 0$ the condition $\delta\omega > 0$ will be definitely fulfilled for $c_2 > 0$. Hence it follows that for $c_2 > 0$ there is a constant increase of ω .

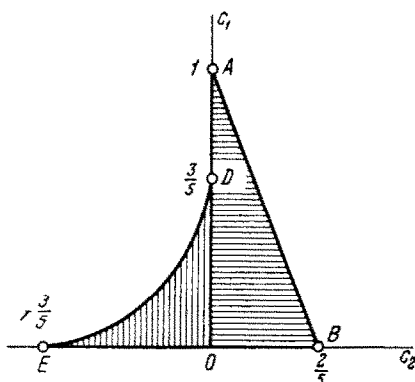


FIG. 1. THE REGION OF POSSIBLE VALUES OF THE CONSTANTS c_1 AND c_2 .

It must be remembered that the condition $c_2 > 0$ corresponds to the class of orbit with the initial value of the longitude of the line of apsides φ_0 satisfying the condition $\sin^2 \varphi_0 = \sin^2 \omega \sin^2 i_0 < \frac{2}{5}$, and arbitrary values of other parameters.

We shall consider the $c_2 \geq 0$. According to (54) and (57) the values $\omega = 0$, $\omega = \frac{\pi}{2}$, $\omega = \pi$ and $\omega = \frac{3\pi}{2}$ are extreme for the functions $\varepsilon(N)$ and $i(N)$. From (57) we have

$$\left(\frac{\delta^2 \varepsilon}{\delta N^2} \right)_{\substack{\omega=0 \\ \omega=\pi}} = -A(1 - \varepsilon) \varepsilon^{\frac{1}{2}} \left[\sin^2 i \frac{\delta}{\delta \omega} (\sin 2\omega) \cdot \frac{\delta \omega}{\delta N} \right]_{\omega=0} < 0,$$

i.e. the points at which $\omega = 0$ or $\omega = \pi$ are points of maximum values of the function ε and consequently, according to (58), points of minimum values of $\cos^2 i$.

At points, where $\omega = \frac{\pi}{2}$ or $\omega = \frac{3}{2}\pi$, a minimum of the function ε and a maximum of $\cos^2 i$ are realized. Substituting the values $\omega = 0$ ($\omega = \pi$) in (59) we shall obtain

$$\varepsilon_{\max} = 1 - \frac{5}{2}c_2. \quad (77)$$

To determine $\varepsilon_{\min}$ from (58) and (59) with $\omega = \frac{\pi}{2}$ ($\omega = \frac{3}{2}\pi$) we obtain the equation

$$\varepsilon_{\min}^2 - \left[1 + \frac{5}{3}(c_1 + c_2)\right]\varepsilon_{\min} + \frac{5}{3}c_1 = 0. \quad (78)$$

Equation (78) has two positive roots. A physically possible solution is determined by the condition $\varepsilon < 1$

$$\varepsilon_{\min} = \frac{1 + \frac{5}{3}(c_1 + c_2) - \sqrt{\left[1 + \frac{5}{3}(c_1 + c_2)\right]^2 - \frac{5}{3} \cdot 4c_1}}{2}. \quad (79)$$

From formula (58) we shall get

$$(\cos^2 i)_{\max} = \frac{c_1}{\varepsilon_{\min}}, (\cos^2 i)_{\min} = \frac{c_1}{\varepsilon_{\max}}. \quad (80)$$

If $\varepsilon_{\min} > \tilde{\varepsilon} = 1 - \left(1 - \frac{R}{a}\right)^2$, where R is the radius of the central body, then in the case considered ($c_2 > 0$) the parameters ε and i fluctuate with period ΔN , equal to the number of revolutions, in the course of which the value of ω changes by π .

$$\Delta N = \frac{1}{A} \int_{\omega_0}^{\omega_0 + \pi} \frac{\varepsilon^2 d\omega}{\left[(\cos^2 i - \varepsilon) \sin^2 \omega + \frac{2}{5}\varepsilon\right]}. \quad (81)$$

If $\varepsilon_{\min} < \tilde{\varepsilon} = 1 - \left(1 - \frac{R}{a}\right)^2$, then for $\varepsilon = \tilde{\varepsilon}$ there will be collision with the central body for values of i and ω , obtained from the integrals (58) and (59).

We shall now consider the case $c_2 < 0$. We shall put $-c_2 = c_2' > 0$. The expression (59) will be written in the form

$$\varepsilon = 1 - \frac{c_2'}{\sin^2 i \sin^2 \omega - \frac{2}{5}}. \quad (82)$$

The condition $c_2' > 0$ requires that $\sin^2 i_0 \sin^2 \omega_0 > \frac{2}{5}$, and consequently also that each of the factors must at least satisfy the condition $\sin^2 \omega_0 > \frac{2}{5}$, $\sin^2 i_0 > \frac{2}{5}$. From (82) it follows that the values of ω will be included in one of the intervals (ω_1^*, ω_3^*) or $(\omega_4^*, \omega_2^* + 2\pi)$ depending on to which of these intervals ω_0 belongs. The extreme values $\varepsilon_{\text{extr}}$ and $(\cos i)_{\text{extr}}$ may be reached in this case only with $\omega = \frac{\pi}{2}$ (or $\omega = \frac{3}{2}\pi$). The corresponding equation for $\varepsilon_{\text{extr}}$ we get from (58) and (82) in the following form:

$$\varepsilon_{\text{extr}}^2 - \left[1 + \frac{5}{3}(c_1 - c_2')\right]\varepsilon_{\text{extr}} + \frac{5}{3}c_1 = 0. \quad (83)$$

Having taken into account the region of possible values of c_1 and c_2 found above, we may show that both roots of equation (83) are positive and lie in the interval between $\varepsilon = 0$ and $\varepsilon = 1$. The limiting cases for the values of $\varepsilon_{\text{extr}}$ correspond to the particular solutions considered in paragraphs 2 and 3 respectively.

The extreme values of ω_{extr} are obtained from the condition $\delta\omega = 0$. From this condition, according to (54) we get

$$\left(\frac{c_1}{(\varepsilon^*)^2} - 1\right) \sin^2 \omega_{\text{extr}} + \frac{2}{5} = 0. \quad (84)$$

Here ε^* is the value of ε with $\omega = \omega_{\text{extr}}$. The equation (84) together with (58) and (82) enables us to obtain the equation for ε^*

$$(\varepsilon^*)^2 \left(c_2' - \frac{2}{5}c_1\right) + \frac{4}{5}c_1\varepsilon^* - c_1\left(\frac{2}{5} + c_2'\right) = 0. \quad (85)$$

It is easy to establish that, for the region considered (c_1, c_2') , in the interval $(\varepsilon = 0, \varepsilon = 1)$ there is only one root for ε^* .

We get two values of ω_{extr} , according to (84), since the two roots are symmetrical with respect to $\omega = \frac{\pi}{2}$ or $\omega = \frac{3}{2}\pi$ (in the case where $\omega_0 \in (\omega_4^*, 2\pi + \omega_2^*)$).

From the above discussion it follows that for $c_2 < 0$ not only the change of ε and i , as was the case with $c_2 > 0$, but also the change of the distance of the pericentre from the node ω bears a periodic character. With this there remains the possibility of the fall of the satellite to the surface of the central body, in the case where the central body has an ultimate radius R and the minimum value ε_{min} satisfying the inequality $\varepsilon_{\text{min}} < \tilde{\varepsilon}$.

The change of position of the node Ω is determined by equation (62) and has the character of a departure, whose direction depends on the sign of $\cos i$.

In Fig. 2 in the plane of extreme values of the eccentricity ($e_{\text{min}}, e_{\text{max}}$) are seen the lines of fixed values of the constant c_1 (continuous lines) and c_2 (dotted lines for values of $c_2 > 0$). In Fig. 3 in the same plane are isolines for c_1 and c_2 for the case $c_2 < 0$. In Fig. 4 for the case $c_2 < 0$ is shown the dependence of extreme values of $\sin^2 \omega_{\text{extr}}$ on c_2 for different values of c_1 . All these graphs were calculated from the formulae cited in the present section.

1. As a result of the discussion, which has taken place in the present section, it has been found that, for the defined region of initial data, the maximum values of the eccentricity e_{max} which are reached in the process of evolution, have a value near to unity or (in the case where $i = 90^\circ$) even of $e_{\text{max}} = 1$. However with values of e , sufficiently near to unity, the value of γ (see paragraph 1 in Section 3) may greatly differ from unity, and the approximate formulae of the present section will not describe the real evolution.

If the central body has an ultimate radius R , the results of the present section concerning the falling of the satellite onto the surface of the central body will be correct (with fulfilment of the rest of the assumptions) in the case where the value of

$$\frac{\mu_k}{\mu} \left(\frac{a}{r_k}\right)^3 \frac{a}{R} \ll 1$$

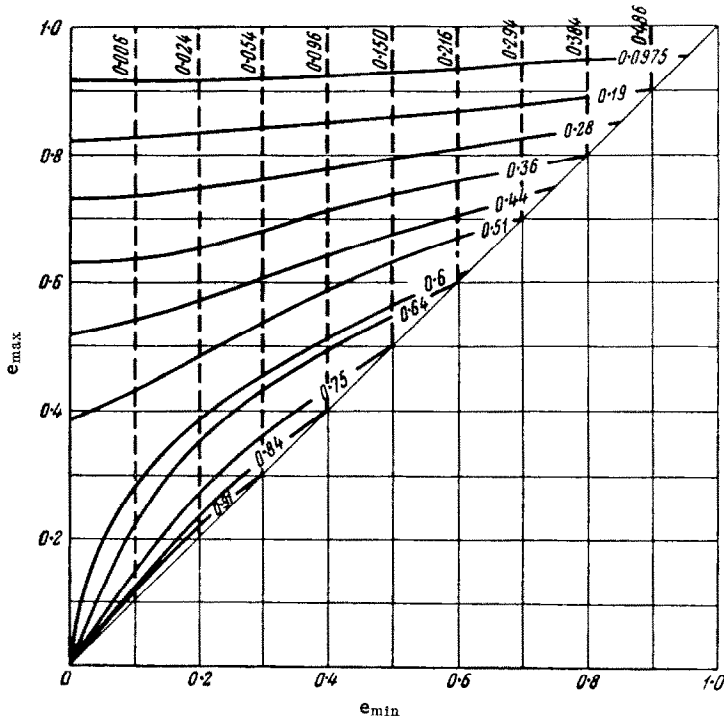


FIG. 2. THE CONNECTION BETWEEN THE EXTREME VALUES OF THE ECCENTRICITY FOR DIFFERENT VALUES OF c_1 (CONTINUOUS LINES) AND $c_2 > 0$ (DOTTED LINES). THE FIGURES ON THE LINES SHOW THE CORRESPONDING VALUES OF THE CONSTANTS c_1 AND $1.5c_2$.

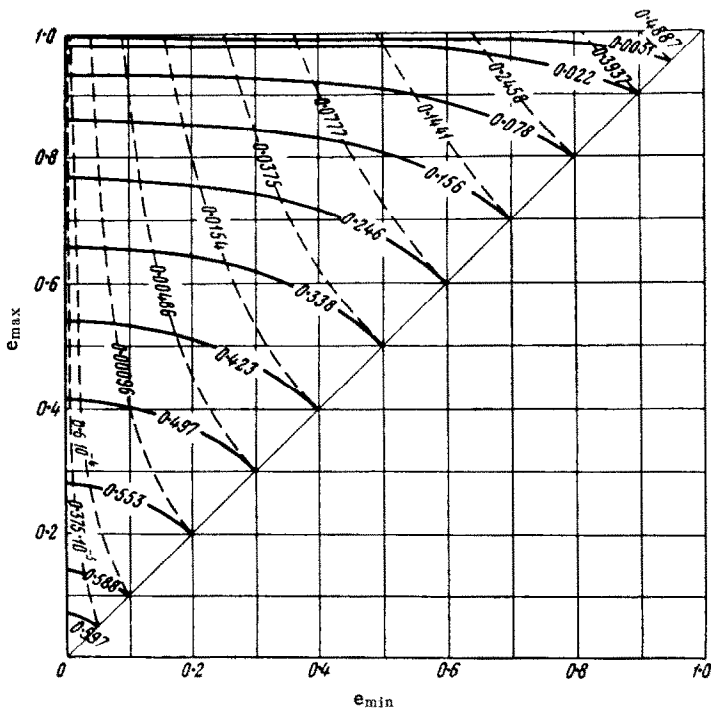


FIG. 3. THE CONNECTION BETWEEN THE EXTREME VALUES OF THE ECCENTRICITY FOR DIFFERENT VALUES OF c_1 (CONTINUOUS LINES) AND $c_2 < 0$ (DOTTED LINES). THE FIGURES ON THE LINES SHOW THE CORRESPONDING VALUES OF THE CONSTANTS c_1 AND $c_2' = -c_2$.

This value characterizes the difference of γ from unity in the worst case of very large values of the eccentricity (from possible ones up to the falling to the surface of the central body).

2. Analysis of the problem discussed in the present section, rests upon the existence of two integrals of the system of equations (54). We note that the first integral $\varepsilon \cos^2 i = c_1$

has a simple physical meaning. As long as $\varepsilon = \frac{p}{a} = \frac{c^2}{\mu a}$, where c is the constant of the angular momentum of the satellite, we have $c \cdot \cos i = \text{const}$, i.e. we get a natural result with the assumptions made: the projection of the vector of angular momentum of the satellite on the normal to the plane of the orbit of the perturbing body, averaged over a revolution of the perturbing body, remains constant in the process of evolution.

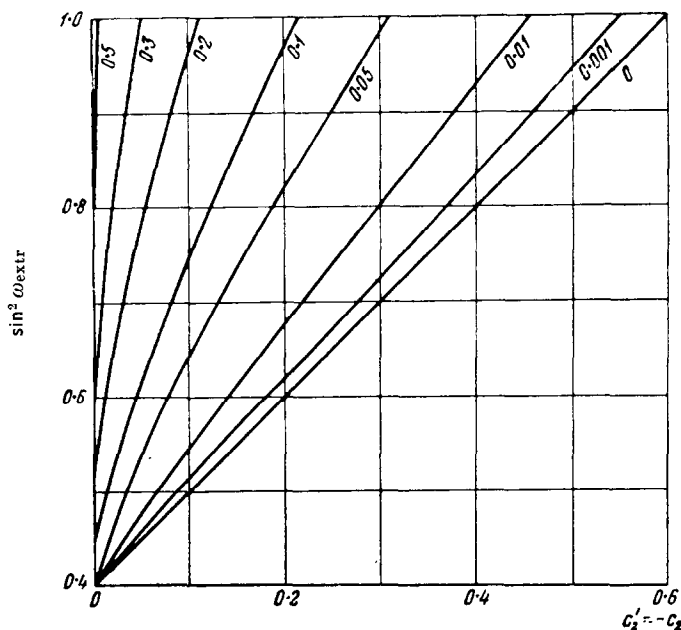


FIG. 4. THE DEPENDENCE OF $\sin^2 \omega_{\text{extr}}$ ON $c_2 < 0$ FOR DIFFERENT VALUES OF c_1 . THE FIGURES NEAR THE LINES SHOW THE CORRESPONDING VALUES OF c_1 .

8. THE VALUE OF THE FLUCTUATIONS OF HEIGHT OF THE PERICENTRE OF A SATELLITE ORBIT IN THE COURSE OF A PERIODIC REVOLUTION OF THE PERTURBING BODY

For analysis of the evolution of the orbit of a satellite with large eccentricity, the investigation of the change of height of the pericentre of the orbit occupies a special place. This is connected with the fact that it often happens to coincide with the cases where the height of the pericentre of the orbit of such a satellite is near to the radius of the central body. In these cases a relatively small change of the angular co-ordinates may be immaterial, but a related change of height of the pericentre of the orbit of such an order will be so considerable that with it there arises a correspondingly large change of distance from the surface of the central body to the pericentre.

We shall work out the value of the variation of height of the pericentre of the satellite's orbit confining ourselves to a period of revolution of the perturbing body, within the limits

of the first approximation $\delta_{11}x$ of the departure of the elements. The change of height of the pericentre δr_π in N revolutions, according to the relation $r_\pi = a(1 - e)$ and the results of Section 6 (with the assumptions of that section) is determined by the formula

$$\delta r_\pi = 15\pi \frac{\mu_k}{\mu} \left(\frac{a}{p_k} \right)^3 a e \varepsilon^3 [\beta_3].$$

From the determination of $[\beta_3]$ from (42) it follows that the extreme values of $[\beta_3]$ and consequently also the extreme values of δr_π are reached with $\beta_3 = \xi_1 \xi_2 \Delta_k^3 = 0$.

We may show that the condition $\xi_1 = 0$ determines the values of the argument of latitude of the perturbing body $u^{\max}$, for which the maxima of the function $[\beta_3]$ are realized and the condition $\xi_2 = 0$ enables us to determine the points of minimum values of this function.

We shall introduce the notation

$$\begin{aligned} \cos \psi_1 &= \frac{d_{11}}{\sqrt{d_{11}^2 + d_{12}^2}}, \quad \sin \psi_1 = \frac{d_{12}}{\sqrt{d_{11}^2 + d_{12}^2}}, \\ \cos \psi_2 &= \frac{d_{21}}{\sqrt{d_{21}^2 + d_{22}^2}}, \quad \sin \psi_2 = \frac{d_{22}}{\sqrt{d_{21}^2 + d_{22}^2}}. \end{aligned}$$

Then from the condition $\xi_1 = 0$ and the expression (24) for ξ_1 we get

$$u_{1,2}^{\max} = \psi_1 \pm \frac{\pi}{2}.$$

Similarly the condition $\xi_2 = 0$ gives the points at which the minima are reached

$$u_{1,2}^{\min} = \psi_2 \pm \frac{\pi}{2}.$$

If for the plane XY we use the plane of the orbit of the perturbing body, the matrix D_1 [formula (26)] takes the form

$$D_1 = \begin{vmatrix} \cos \omega \cos \Omega - \sin \omega \sin \Omega \cos i & \cos \omega \sin \Omega + \sin \omega \cos \Omega \cos i \\ -\sin \omega \cos \Omega - \cos \omega \sin \Omega \cos i & -\sin \omega \sin \Omega + \cos \omega \cos \Omega \cos i \\ \sin \Omega \sin i & -\cos \Omega \sin i \end{vmatrix}.$$

Hence we get

$$\begin{aligned} d_{11}^2 + d_{12}^2 &= \cos^2 \omega + \sin^2 \omega \cos^2 i = \cos^2 \varphi, \\ d_{21}^2 + d_{22}^2 &= 1 - \cos^2 \omega \sin^2 i, \\ d_{11}d_{22} - d_{21}d_{12} &= \cos i. \end{aligned}$$

With the help of these relations we find that

$$\sin(\psi_2 - \psi_1) = \frac{\cos i}{\cos \varphi \sqrt{1 - \sin^2 i \cos^2 \omega}}.$$

With $\sin i = 0$ (the plane of the orbit of the satellite coincides with the plane of the orbit of the perturbing body) $\psi_2 - \psi_1 = \frac{\pi}{2}$ and, consequently the radius vectors of the perturbing body $\mathbf{r}_k(u^{\min})$ and $\mathbf{r}_k(u^{\max})$ are orthogonal. With $i = \frac{\pi}{2}$ we have $\psi_2 = \psi_1$ and consequently the extreme points are points of inflection of the function.

The orthogonality of the vectors $\mathbf{r}_k(u^{\min})$ and $\mathbf{r}_k(u^{\max})$ except for the case $\sin i = 0$, will take place with $\sin 2\omega = 0$, i.e. when there are realized the cases of symmetrical and anti-symmetrical position of the satellite's orbit relative to the plane of the orbit of the perturbing body. In the general case $|u_1^{\max} - u_1^{\min}| \neq |u_2^{\max} - u_1^{\min}|$ although it is always true that $|u_2^{\max} - u_2^{\min}| = |u_1^{\max} - u_1^{\min}|$ and $|u_2^{\max} - u_1^{\max}| = \pi$, $|u_2^{\min} - u_1^{\min}| = \pi$.

In expression (46) we shall substitute for the components of the vector $\mathbf{V}_2$ the quantities $u_N = u_{1,2}^{\max}$ and $u_N = u_{1,2}^{\min}$ and in accordance with (42) we shall determine $[\beta_3]$ at these points. We shall assume for the simplification of the formula, that the X axis is directed towards the pericentre of the orbit of the perturbing body ($\omega_k = 0$). In this case the formulae for $[\beta_3]$ at the given points will have the form

$$[\beta_3(u_{1,2}^{\max})] = \left[\beta_3 \left(\psi_1 \pm \frac{\pi}{2} \right) \right] = \frac{1}{bT} \left\{ - \frac{\sin^2 i \sin 2\omega}{2} \cdot \frac{\psi_1 \pm \frac{\pi}{2}}{2} + \frac{1}{2} d_{11} d_{22} \right. \\ \left. \pm e_k \frac{\frac{2}{3} d_{11}^4 d_{21} + \frac{1}{3} d_{11}^3 d_{12} d_{22} + d_{11}^2 d_{12}^2 d_{21} + d_{11} d_{12}^3 d_{22} + d_{12}^4 d_{21}}{(d_{11}^2 + d_{12}^2)^{3/2}} \right\} + C, \\ [\beta_3(u_{1,2}^{\min})] = \left[\beta_3 \left(\psi_2 \pm \frac{\pi}{2} \right) \right] = \frac{1}{bT} \left\{ - \frac{\sin^2 i \sin 2\omega}{2} \cdot \frac{\psi_2 \pm \frac{\pi}{2}}{2} + \frac{1}{2} d_{21} d_{12} \right. \\ \left. \pm e_k \cdot \frac{\frac{2}{3} d_{21}^4 d_{11} + \frac{1}{3} d_{21}^3 d_{22} d_{12} + d_{21}^2 d_{22}^2 d_{11} + d_{21} d_{22}^3 d_{12} + d_{22}^4 d_{11}}{(d_{21}^2 + d_{22}^2)^{3/2}} \right\} + C,$$

where $C = [\beta_3(u_0)]$ is a constant determined by the initial conditions.

The change δr_π from maximum to minimum value (in a sense the amplitude of the vibrations) will be determined by the formula

$$\mathcal{D} = \delta r_\pi^{\max} - \delta r_\pi^{\min} = 15\pi \frac{\mu_k}{\mu} \left(\frac{a}{p_k} \right)^3 a e \varepsilon^{\frac{1}{2}} \left\{ [\beta_3(u^{\max})] - [\beta_3(u^{\min})] \right\}.$$

The value of $\mathcal{D}$ may be given in the form of three items $\mathcal{D} = \mathcal{D}_1 + \mathcal{D}_2 + \mathcal{D}_3$ where $\mathcal{D}_1$ answers the first term standing in the braces of the formula for $[\beta_3]$. The increment $\mathcal{D}_1$ is connected with the presence of a secular departure of the height of the pericentre for a revolution of the perturbing body. The component $\mathcal{D}_2$ is connected with the second terms in the braces and is equal to

$$\mathcal{D}_2 = 15\pi \frac{\mu_k}{\mu} \left(\frac{a}{p_k} \right)^3 a e \varepsilon^{\frac{1}{2}} \frac{\cos i}{2} \frac{1}{bT}.$$

This component determines the amplitude of the fluctuations δr_π with the absence of a secular departure of the elements for a revolution of the perturbing body and for $e_k = 0$ (circular orbit of the perturbing body).

Item $\mathcal{D}_3$ answers to terms proportional to the eccentricity of the orbit of the perturbing body.

With the passage from the first maximum $u_1^{\max}$ to the first minimum $u_1^{\min}$ the component $\mathcal{D}_2$ is equal in magnitude and opposite in sign to the value of $\mathcal{D}_2$ answering to the passage

from the first minimum to the second maximum $u_3^{\max}$. Hence it follows that, for circular orbits of the perturbing body and with the absence of a secular departure of the elements for a revolution of the perturbing body, the height of the pericentre will twice reach the maximum values (equal to one another) and twice reach the minimum (equal to one another) in the course of a revolution of the perturbing body. Moreover with the absence of a secular departure for a revolution of the perturbing body, if its orbit has eccentricity differing from zero, two maximum values of δr_π , which are reached in the course of one revolution of the perturbing body, in the general case will no longer be equal in magnitude. Also in the general case the two minimum values of δr_π will not be equal.

With the assumptions of sections 4–6 it is easy to obtain formulae for the extreme values also of all the remaining elements of the orbit.

For this, just as it was done above for the increment δr_π , first we determine the value of the argument of the latitude of the perturbing body $u_{x_i}^{\text{extr}}$ for which the extreme of the given element x_i is realised. The value of $u_{x_i}^{\text{extr}}$ is found from the condition $\Delta x_i = 0$ where Δx_i is determined by the relations (17). Substituting the values $u_{x_i}^{\text{extr}}$ in the expression for δx_i (formulae (17) with β_i changed to $[\beta_i]$) we find the value we seek.

In these operations using the following approximations for the increments ($\Delta_{12}x$, $\Delta_{22}x$, ... and $\delta_{12}x$, $\delta_{22}x$, ...), within the limits of the assumptions of the present work, we may make more exact the magnitudes of the extreme values of the elements of the satellite's orbit for a revolution of the perturbing body.

9. THE METHOD OF CALCULATION OF THE EVOLUTION OF THE ORBIT OF AN ARTIFICIAL EARTH SATELLITE BY THE APPROXIMATE FORMULAE

For artificial earth satellites with large heights of apogee, in the general case basic factors determining the evolution of the orbit will be: (1) the perturbing influence of the Moon; (2) the perturbing influence of the Sun; (3) the difference between the gravitational field of the Earth and the central; (4) the braking in the Earth's atmosphere.

For calculating the secular departure of the elements of the orbit of an Earth satellite for one revolution on account of the difference between the gravitational field of the Earth and the central, we may use the formulae obtained in⁽⁶⁾

$$\left. \begin{aligned} \Delta_c \Omega &= -\frac{2\pi\alpha}{p^2\mu} \cos i, \\ \Delta_c \omega &= \frac{\pi\alpha}{p^2\mu} (5 \cos^2 i - 1), \end{aligned} \right\} \quad (86)$$

where $\alpha = 0.266360022 \cdot 10^{11} \text{ km}^5 \cdot \text{sec}^{-2}$. Here Ω , ω , i are relative to the plane of the Earth's equator.

The approximate formulae are obtained by integrating the respective equations in the osculating elements with assumptions, equivalent to the assumptions noted in paragraphs 1 and 2 of Section 3 of the present work, and with the calculation of a first approximation in the expansion for the gravitational potential of the Earth. The remaining elements of the orbit in the considered approximation have no secular departure connected with the difference between the gravitational field of the Earth and the central.

According to the second of equations (54), the average changes of eccentricity (and consequently also the height of the orbit's perigee) due to the perturbation of the planets, are proportional to $\sin^2 i \cdot \sin 2\omega$, where ω and i are measured from the plane of the orbit of the perturbing body.

For investigation of the combined influences of the perturbations of the Moon, Sun and the difference between the gravitational field of the Earth and the central, for a long period of time the non-centrality of the gravitational field will show an important effect, in particular, also on the character of the change of height of perigee. Also besides the straight-forward change of $\sin 2\omega$ because of the displacement of the latitude of perigee $\Delta_c\omega$, there will be a change of the parameter $\sin^2 i \cdot \sin 2\omega$, on account of the turning of the satellite's orbit through an angle $\Delta_c\Omega$ around the pole of the Earth. On account of the existing inclination of the plane of the lunar orbit and the plane of the ecliptic to the Earth's equator with such turning there will also be a change of i and ω .

Within the limits of the approximate discussion, for calculation of the change, during a revolution of the satellite, of the semi-axis ($\Delta_T a$) and eccentricity ($\Delta_T e$) on account of the braking of the satellite in the atmosphere, the last formulae (e.g. formulae (5)–(7) of work⁽⁹⁾) may also be used. However for many investigations of the problem we may leave the effect of the braking of the atmosphere on the evolution of the orbit, limiting ourselves to the examination of orbits which have perigee further from the surface of the Earth, e.g. more than 300 km. In this case the braking of the satellite in the atmosphere will often have little influence on the evolution of the orbit in comparison with other factors.

As the basic plane XY for the measurement of the angles we may use the plane of the Earth's equator. In the calculation, as long as it is assumed that we are also calculating the evolution caused by the non-centrality of the Earth's field ($\Delta_c\Omega$ and $\Delta_c\omega$) the plane of the Moon's orbit or the ecliptic will have no advantages.

For approximate investigation we may consider that the Moon and Sun move around the Earth in ellipses, with given parameters $p_1, e_1 \dots$ for the Moon and $p_2, e_2 \dots$ for the Sun. The positions of the Moon and Sun in their orbits are given at the initial moment by the arguments of latitude u_1 and u_2 respectively.

Besides the parameters of the orbits and the constants of gravitational potential of the Moon μ_1 and Sun μ_2 the periods of revolution of the Moon T_1 and Sun T_2 around the Earth are given. Here obviously it is advisable to use the sidereal periods in the calculation.

The increments Δx_s of the element x_s of the satellite's orbit for a revolution is determined as the sum of the increments

$$\Delta x_s = \Delta_{11}^{(1)} x_s + \Delta_{12}^{(1)} x_s + \Delta_{21}^{(1)} x_s + \Delta_{11}^{(2)} x_s + \Delta_{12}^{(2)} x_s + \Delta_{21}^{(2)} x_s + \Delta_c x_s, \quad (87)$$

$$s = 1, 2, 3, 4, 5.$$

The upper indices in $\Delta_{ij}^{(k)} x$ correspond to the increments connected with the action of Moon (index 1) and Sun (index 2). The lower indices, as adopted in this work, correspond to the first and second approximations (formulae (4)–(5)) in the expansion of the perturbing acceleration (index i), and to the first and second approximations (formula (11)) for the calculation of the motion of the perturbing body in the course of one revolution of the satellite (index j). The term $\Delta_c x_s$ denotes the increment caused by the difference between the gravitational field of the Earth and the central.

If the satellite's perigee is found close to the Earth's surface, and the height of apogee $r_x \approx 50 \cdot 10^3$ km the quantity $\frac{r_x}{p_1} \approx \frac{1}{7}$. For the calculation, in the expansion of the perturbing acceleration, of terms of the order $O\left(\frac{r_x^2}{p_1^2}\right)$ (i.e. of the increments $\Delta_{21}^{(1)} x_3$) we may expect that the error in the calculation of the secular changes of the elements for a revolution will be of

the order of a few per cent. For such a satellite there may be errors of that order in the calculation of only a first approximation in the changes of the elements on account of the motion of the Moon in the process of one revolution $\Delta_{12}x_s$. Using an accuracy of several per cent it is permissible, perhaps, for example, not to take into account in the calculation the increments of the elements $\Delta_{12}^{(2)}x_s$ and $\Delta_{21}^{(2)}x_s$, connected with the influence of the Sun.

To calculate the increments $\Delta_{ij}x_s$ (or $\delta_{ij}x_s$) it is necessary to determine the values of the arguments of latitude $u_1^{(m+1)}$ and $u_2^{(m+1)}$ at the end of the step (with respect to the time) in which the increments are determined.

From the given elements of the orbits of the perturbing bodies, known $u_k^{(m)}$ at the beginning of the m -th step and the given step $t^{(m+1)} - t^{(m)}$ we can determine the values $u_k^{(m+1)}$ ($k = 1, 2$). For this, generally speaking, we have to solve Kepler's Equation. As long as the orbits of the perturbing bodies have small eccentricities, we may use various approximate methods for the solution of this equation⁽¹⁰⁾.

If the increments for one revolution are considered then $t^{(m+1)} - t^{(m)} = T$, where T is the period of the satellite. For the calculation of the increments $\Delta_{ij}x_s$ for one revolution of the satellite, with sufficiently small eccentricity of the orbit of the perturbing body e_k and small $\frac{T}{T_k}$, the increment of the argument of latitude for a revolution $\Delta u_k = u_k^{(m+1)} - u_k^{(m)}$ may be obtained as a first approximation in the form

$$\Delta u_k = \frac{2\pi \Delta_k^2 T}{\varepsilon_k^{\frac{3}{2}} T_k}, \quad (88)$$

and the quantity u_k^* , which one must use for the calculation of $\alpha^*, \beta^*, \gamma^*, \left(\frac{d\beta}{dt}\right)^*$ according to formulae (36), (39) is obtained in the form

$$u_k^* = u_k^{(m)} + \frac{\Delta u_k}{2}. \quad (89)$$

If the speed of change of the elements of the satellite's orbit allows us to consider the increments of the elements straight away for n revolutions, we find the resulting increment similarly to (87) in the form of a sum,

$$\delta x_s = \delta_{11}^{(1)}x_s + \dots + \delta_{21}^{(2)}x_s + \delta_c x_s \quad (s = 1, 2, \dots, 5). \quad (90)$$

The increments $\delta_{ij}x_s$ are determined according to (17)–(19) with the use of vectors $[\alpha]$, $[\beta]$, $[\gamma]$, $\left[\frac{d\beta}{dt}\right]$ in the form (42). The values of $u_k^{(m+1)}$ ($k = 1, 2$) are found by the solution of Kepler's Equation, with the increment of time equal to $t^{(m+1)} - t^{(m)} = nT$.

For departures $\delta_c x_s$ due to the non-centrality of the gravitational field we have $\delta_c x_s = n \Delta_c x_s$ where $\Delta_c x_s$ is determined by the relations (86).

The system of equations (87) or (89) is a closed system of difference equations for the dependent variables $x_1, \dots, x_5$,

$$\left. \begin{aligned} \delta x_s &= f_s(x_1, \dots, x_5, N) \quad (s = 1, 2, \dots, 5) \\ \Delta x_s &= \Phi_s(x_2, \dots, x_5, N) \quad (s = 1, 2, \dots, 5). \end{aligned} \right\} \quad (91)$$

For the independent variable we may, for example, use the ordinal number of the satellite's revolution N .

We may use the system (91) for the calculation of the evolution of the satellite's orbit for a prolonged time. For this it is necessary for the system (91) to solve Cauchy's problem.

In the simplest case we may use Euler's method. If the known elements $x_s^{(m)}$ are in the m -th step, the elements in the $(m+1)$ th step are defined as $N^{(m+1)} = N^{(m)} + 1$, $x_s^{(m+1)} = x_s^{(m)} + \Delta x_s^{(m)}$ or for a step of n revolutions $N^{(m+1)} = N^{(m)} + n$, $x_s^{(m+1)} = x_s^{(m)} + \delta x_s^{(m)}$. Obviously we may always use when necessary Euler's method of evaluation in which, after the determination of $\Delta x_s^{(m)}$ from the values of $x_s^{(m)}$ in the initial step, the increments Δx_s are evaluated by using on the right-hand sides of (91) the average value of x_s over one step.

$$\Delta x_s^{(m+\frac{1}{2})} = \Phi_s \left(x_1^{(m)} + \frac{\Delta x_1^{(m)}}{2}, x_2^{(m)} + \frac{\Delta x_2^{(m)}}{2}, \dots, N \right).$$

These very simple methods may always be used. The possibility and the advisability of using for system (91) methods of solution of higher order⁽⁷⁾ resting on the ease with which these solutions for x_s may be found, must be specially investigated.

The averaged increments $\frac{\delta x}{\delta N}$ according to (54) essentially depend on the values of ω and i , being measured from the plane of the orbit of the perturbing body. Besides the factors shown above, causing the change of these elements, we should note the influence of the solar precession of the Moon's orbit on the values of ω and i which are measured from the plane of the lunar orbit. The node of the plane of the Moon's orbit on the ecliptic on an average changes by 0.05 degrees in 24 hr. For calculating the evolution over a long period of time this effect necessitates substantial corrections in the magnitudes of the departures of the parameters of the satellite's orbit. For the calculation of the evolution over a short period of time, without changing the methods of calculation resting upon an unchanging orbit of the perturbing body, we may introduce at each step corrections to the elements of the Moon's orbit, making allowance for its precession. Thus we may take into account parametrically the change, of angular distance of the perigee of the Moon's orbit from the equatorial node, brought about by the same effect.

10. COMPARISON OF THE RESULTS OF CALCULATION BY THE APPROXIMATE FORMULAE WITH THE ACCURATE SOLUTION OF THE PROBLEM BY NUMERICAL INTEGRATION OF DIFFERENTIAL EQUATIONS

The sum total of the estimates of the accuracy of the solution of a problem by the approximate formulae was found by comparing the calculations by the formulae of the present work with the numerical integration of the system of differential equations of celestial mechanics. Examples on the evolution of the parameters of the orbits of Earth and Moon satellites were calculated. The problem of the combined motion of the Earth, Moon, Sun and satellite was solved accurately. Integration was carried out in a non-rotating, Cartesian system of co-ordinates, connected with the centre of the central body. For a series of consecutive rotations, at the minimum distance of the orbit from the central body, the osculating elements were calculated with the Cartesian co-ordinates. The change of these elements was also compared with the corresponding values, calculated by the approximate formulae.

The errors of the approximate method will be the greater, the greater the maximum distance of the satellite from the central body. Therefore, for the purpose of comparing the values, orbits of satellites were adopted with apocentre sufficiently far from the central body. For this they were limited by measurements of orbits of satellites for which the accuracy of the approximate method would be still satisfactory for the aims of investigation. From the degree of decrease and increase of error in the measurement of the orbit of the satellite

reliable ideas about the dependence (according to the order of magnitude) of the error of the approximate method on the parameters of the satellite's orbit may be obtained.

In comparison there were considered orbits of Earth satellites with the type of orbit of Explorer VI with heights of perigee of the order of the Earth's radius and heights of apogee of the order of $50-70 \cdot 10^3$ km. In these cases the error of the approximate calculation constituted less than 5 per cent of the value of the departure. For the orbits considered we may expect an error of such an order on the whole because of the use of only first and second approximations in the expansion of the perturbing acceleration due to the Moon in the approximate calculation.

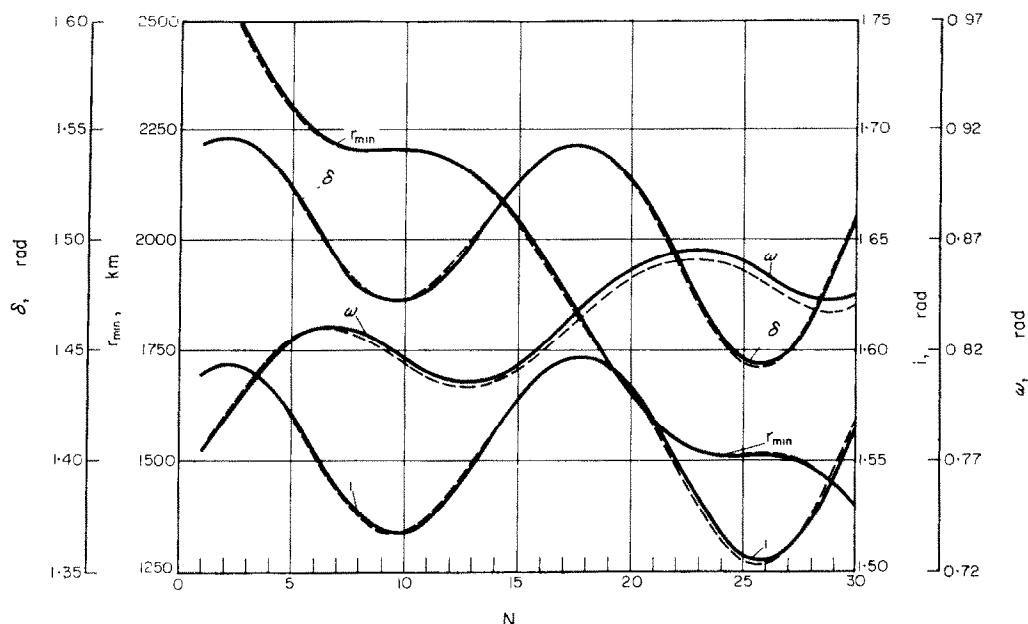


FIG. 5. THE DEPENDENCE OF THE OSCULATING ELEMENTS OF THE ORBIT OF THE MOON SATELLITE ON THE ORDINAL NUMBER OF THE REVOLUTION (VARIANT NO 1).

The continuous curves—the accurate integration of a system of differential equations for the combined motion of the Earth, Moon, Sun and satellite; the dotted curves—the calculation by the approximate formulae.

In Figs. 5–9 for a Moon satellite graphs were introduced showing the dependence of the osculating elements of the satellite's orbit on the number of the revolution at minimum distance from the Moon. The dependence was obtained by numerical integration of the differential equations (continuous lines). For comparison the same dependences were introduced, after being obtained by the approximate formulae (dotted lines). The approximate calculations were also brought into step, in one revolution, with the use for the integration of system (91) of the method of Euler for the calculation.

The initial minimum and maximum distances from the Moon for five calculated variants are given in Table 1.

The angular elements in the calculations and on the graphs are measured from the plane of the Earth's equator. The first two variants (Figs. 5 and 6) are associated with the orbit of a satellite with inclination to the plane of the Moon's orbit near 90° . Here we evaluate the

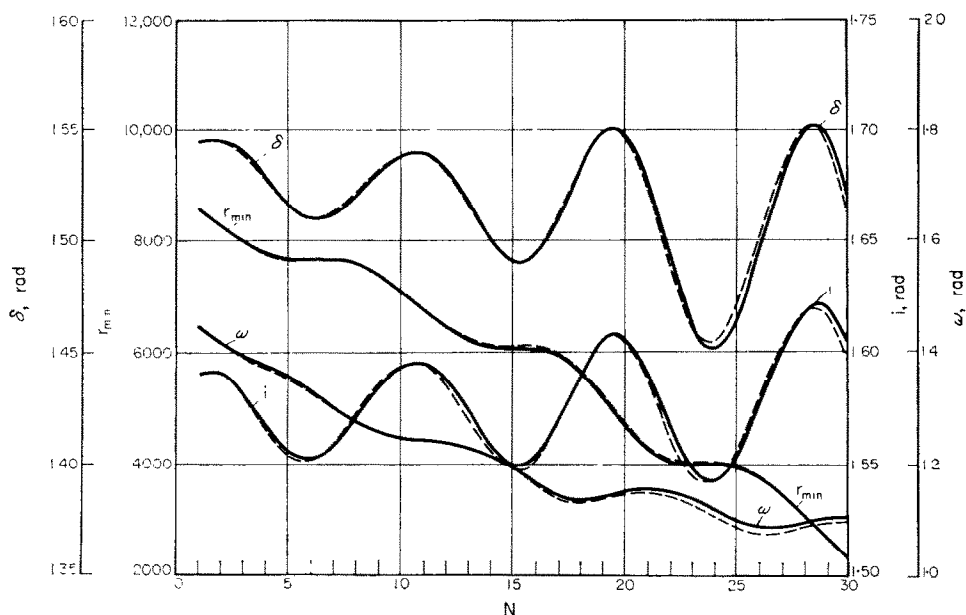


FIG. 6. THE DEPENDENCE OF THE OSCULATING ELEMENTS OF THE ORBIT OF THE MOON SATELLITE ON THE ORDINAL NUMBER OF THE REVOLUTION (VARIANT NO 2). NOTATION THE SAME AS IN FIG. 5.

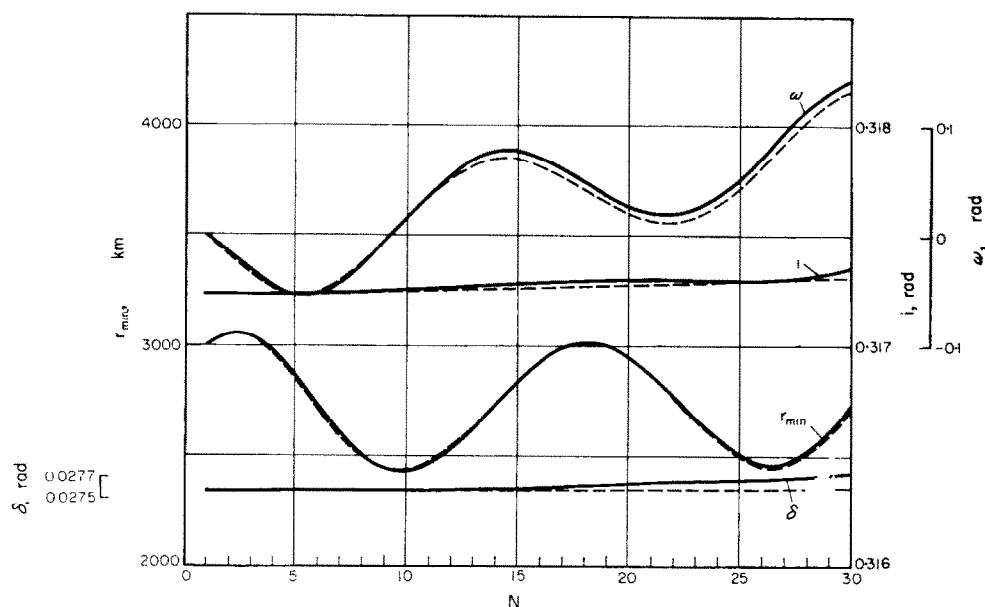


FIG. 7. THE DEPENDENCE OF THE OSCULATING ELEMENTS OF THE ORBIT OF THE MOON SATELLITE ON THE ORDINAL NUMBER OF THE REVOLUTION (VARIANT NO 3).

The continuous curves—the accurate integration of a system of differential equations for the combined motion of the Earth, Moon, Sun and satellite; the dotted curves—the calculation by the approximate formulae.

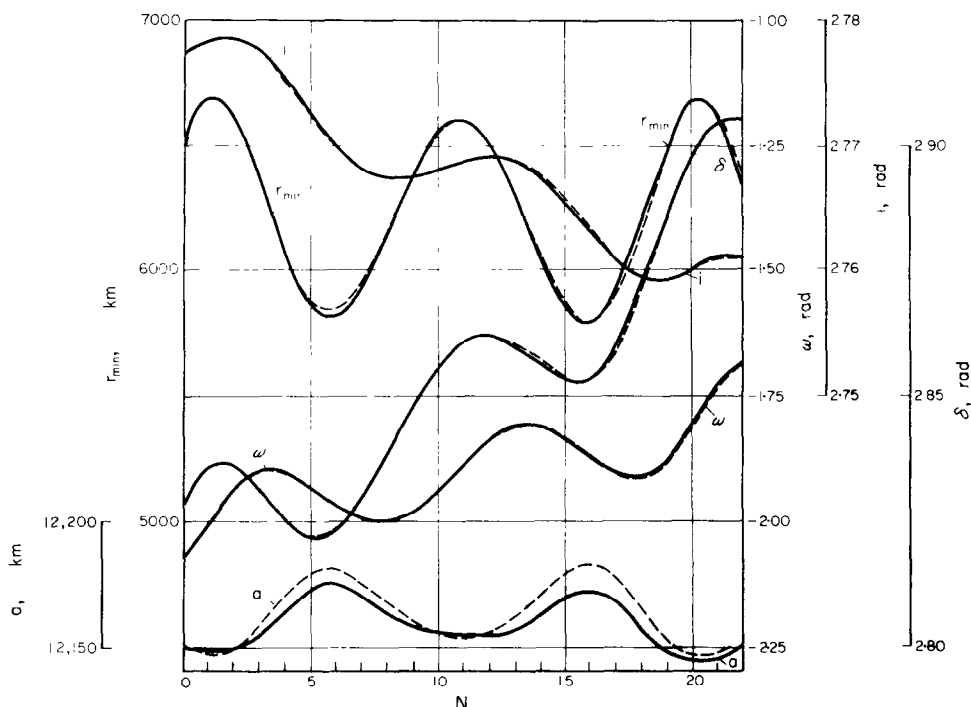


FIG. 8. THE DEPENDENCE OF THE OSCULATING ELEMENTS OF THE ORBIT OF THE MOON SATELLITE ON THE ORDINAL NUMBER OF THE REVOLUTION (VARIANT NO 4). The continuous curves—the accurate integration of a system of differential equations for the combined motion of the Earth, Moon, Sun and satellite; the dotted curves—the calculation by the approximate formulae.

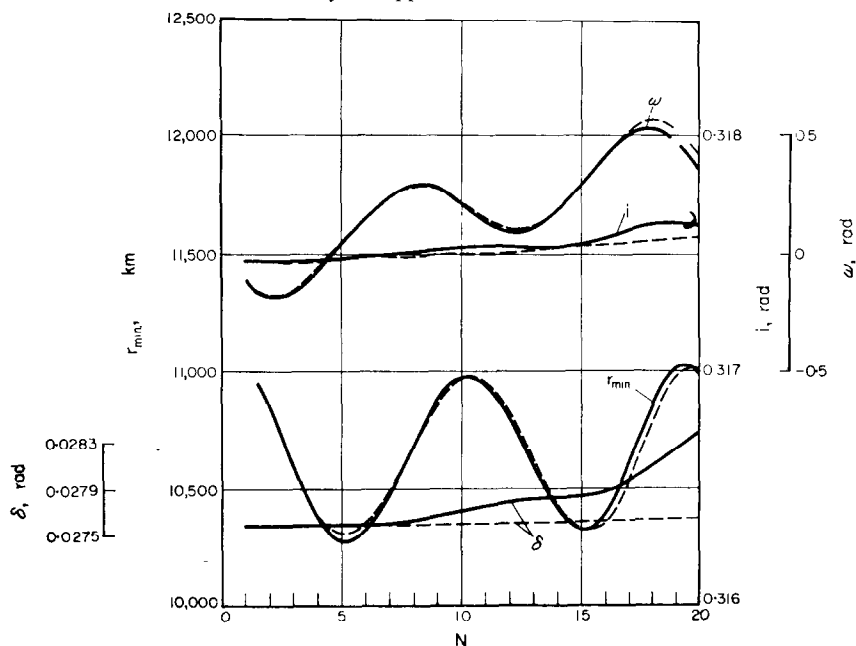


FIG. 9. THE DEPENDENCE OF THE OSCULATING ELEMENTS OF THE ORBIT OF THE MOON SATELLITE ON THE ORDINAL NUMBER OF THE REVOLUTION (VARIANT NO 5). The continuous curves—the accurate integration of a system of differential equations for the combined motion of the Earth, Moon, Sun and satellite; the dotted curves—the calculation by the approximate formulae.

basic case, where the height of pericentre has, on the whole, a monotonic departure in the case of a full rotation of the Earth. The graphs (Figs. 7-9) are associated with orbits, whose planes were near the plane of the orbit of the perturbing body (Earth). In this case the height of the pericentre has fluctuations with period approximately equal to half the lunar month. A comparison is made for the space of 20-30 rev of the satellite. For all the orbits considered this number of revolutions is equivalent to a time greater than a lunar month.

TABLE 1

Variant	$r_{\max}$, 10^3 km	$r_{\min}$, 10^3 km
1	15.35	2.65
2	17.34	8.54
3	15.00	3.00
4	17.80	6.50
5	26.00	11.00

For the orbits of the Moon satellite which were examined, a basic factor, determining the errors of the approximate method, must be the large magnitude of the perturbing acceleration (the condition of unchanging elements of the orbit in the course of one revolution being broken) and the long periods of revolution (errors bound up with the calculation of only a first approximation in the motion of the perturbing body in the course of a revolution of the satellite). On the basis of the discovered values we may conclude that for the aims of investigation and the aims of preliminary calculations the approximate method may be employed for a sufficiently wide class of satellite orbit (in the sense of cosmic flights in practice).

In conclusion I wish to make use of this opportunity to thank D. E. Okhotsimskii for the series of values used in connection with this present paper.

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